ATI LIMITED

Payroll Management System

Business Requirements Document (BRD)

Tier of employee setup, salary heads, payroll run, bonus, bank and cash payslips, and salary certificate with HR approval. Designed as independent modules (low coupling, high cohesion) and integrated with the ERP employee master.

Item	Detail
Document ID	ATI-PAYROLL-BRD-V2.0
Version	2.0 (Draft for review)
Date	27 October 2026
Prepared by	ATI Limited, Business Analysis
Phase	Phase 1 (Phase 2 scope listed in Chapter 18)
Source documents	Reviewed BRD v1.1 with reviewer comments; Payroll Software Requirements Collection (working draft); 01-Payroll_Software_2026.xlsx (sheets New Calculation and Sheet1); Payroll Software Technical Document v0.1; business clarifications of 27 October 2026
Status	Items marked Proposed need business confirmation; see Chapter 19

Document Control

Revision History

Version	Date	Author	Description
0.1	24 Oct 2026	Business Analysis	Technical document: process, business rules, integration, dependencies, Phase 2 scope.
1.0	24 Oct 2026	Business Analysis	First BRD: module-wise business logic, screen specifications, process flows, acceptance criteria, open items.
1.1	24 Oct 2026	Business Analysis	ERP integration moved from a module into the Dependencies chapter as an API call; modules and chapters renumbered.
2.0	27 Oct 2026	Business Analysis	Modules refined as per review comments: Designation Setup replaced by Tier of Employee Setup; salary head and parameter setup placed under Payroll Management and bonus setup under Bonus Management; Bank and Cash payslips and bonus brought into Phase 1; salary certificate split into ERP request, HR approval and issuance. Salary increment as a separate input on the payroll profile. Features, consolidated business logic, screens, user stories, role and actor mapping and process flows added for every module.

Summary of Changes from v1.1

Area	v1.1	v2.0 (this document)
Designation	Module 1 Designation Setup: a typed list of designations linked to a scale.	Module 1 Tier of Employee Setup. The employee designation is fetched from the ERP and mapped to a tier; the employee-to-scale mapping is held in the payroll profile.
Setup ownership	Module 3 Percentage Setup held H/R, M/A, conveyance, bank target and bonus % in one screen.	Salary heads and payroll parameters sit inside Module 3 Payroll Management. Bonus percentages sit inside Module 4 Bonus Management.
Salary heads	Fixed list: H/R, M/A, Conveyance.	Heads (H/R, M/A, Conveyance, etc.) are setup records; new heads can be added without code change.
Bonus	60% of Total, Bank and Cash salary of a finalized payroll run.	Bonus Type = name with percentage. Bonus runs on gross salary at any time of the year, independent of the payroll run. Bank bonus is 80% of the bonus; the rest is cash bonus.
Payslip	Phase 2.	Phase 1, Module 5: Bank and Cash payslips generated, printed and stored with each finalized salary run.
Salary increment	Not covered; Total Salary was overwritten by a new version.	Module 3 (profile): the increment is entered in a separate input and added to the current Total Salary; the new Total Salary drives tuning, payroll and bonus from its effective date.
Salary certificate	One module: HR requests, Management approves.	Employee requests in the ERP; HR approves in the Approval Desk (Module 6); Module 7 issues the certificate. Employee downloads from the ERP; HR can download from Payroll, including a working copy before approval.
Business logic	41 rules (BR-01 to BR-41) in one flat list.	35 consolidated rules (BL-01 to BL-35) grouped by theme; each shows the v1.1 rules it merges.
Module structure	Purpose, logic and screens.	Purpose, features with functionality, business logic, process flow, UI screens, validations and user stories for every module, plus architecture, menu, role and actor map.
Menu	Not described.	Attendance, leave and overtime appear as disabled Phase 2 placeholders.

Review and Approval

Role	Name	Signature	Date
Business Owner / Management			
HR Administrator			
Finance / Accounts			
Payroll Administrator			
ERP / IT Representative			
ATI Business Analyst			

Contents

Document Control **2**
Revision History 2
Summary of Changes from v1.1 2
Review and Approval 3

1. Executive Summary **6**

2. Introduction **7**
2.1 Purpose 7
2.2 Background: How Payroll Works Today 7
2.3 Business Objectives 7
2.4 Scope 7
2.5 Stakeholders 8
2.6 Glossary 8

3. Overall Description **10**
3.1 End-to-End Process Flow 10
3.2 Modular Architecture and Design Principles 10
3.3 Module Overview 12
3.4 Application Menu Structure 13
3.5 Roles, Actors and Permissions 13
3.6 Assumptions and Constraints 15

4. Business Logic (Consolidated) **16**
4.1 Formula Sheet 16
4.2 Business Rules Catalogue 16
4.3 Calculation Reference 21

5. Module 1: Tier of Employee Setup **23**
5.1 Purpose 23
5.2 Features 23
5.3 Business Logic 23
5.4 UI Screen Specification 23
5.5 User Stories 24

6. Module 2: Scale Setup **26**
6.1 Purpose 26
6.2 Features 26
6.3 Business Logic 26
6.4 UI Screen Specification 26
6.5 User Stories 27

7. Module 3: Payroll Management **28**
7.1 Purpose 28
7.2 Features 28
7.3 Business Logic 29
7.4 Process Flow 31
7.5 UI Screen Specification 33
7.6 Validations and Exceptions 37
7.7 User Stories 38

8. Module 4: Bonus Management **40**
8.1 Purpose 40
8.2 Features 40
8.3 Business Logic 40
8.4 Process Flow 41
8.5 UI Screen Specification 42
8.6 User Stories 43

9. Module 5: Payslip Management **44**
9.1 Purpose 44
9.2 Features 44
9.3 Business Logic 44

9.4 Process Flow 46
9.5 UI Screen Specification 47
9.6 User Stories 48

10. Module 6: Salary Certificate Request Approval 49
10.1 Purpose 49
10.2 Features 49
10.3 Business Logic 49
10.4 Process Flow 50
10.5 UI Screen Specification 50
10.6 User Stories 52

11. Module 7: Salary Certificate Issuance 53
11.1 Purpose 53
11.2 Features 53
11.3 Business Logic 53
11.4 Process Flow 54
11.5 UI Screen Specification 55
11.6 User Stories 57

12. Module 8: Reports 58
12.1 Purpose 58
12.2 Features 58
12.3 Report Catalogue 58
12.4 Report Rules 59
12.5 User Stories 59

13. User Story Map: Role and Actor 60
13.1 Stories by Role 60
13.2 Stories by Actor 60
13.3 Stories by Module 61

14. System-Wide Controls (Common Services) 62
14.1 Common Services 62
14.2 Audit Trail 62
14.3 Security and Privacy 62
14.4 Data Integrity 62
14.5 Non-Functional Requirements 62

15. Dependencies 64
15.1 ERP Integration (API): Payroll calls the ERP 64
15.2 Payroll Interface: the ERP calls Payroll 64
15.3 Other Dependencies 64

16. Data Migration and Baseline Reconciliation 65
16.1 Migration Steps 65
16.2 Baseline Totals (18 Employees) 65

17. Acceptance Criteria 66

18. Phase 2: Future Scope 68

19. Open Items and Decisions Required 69

20. Conclusion and Sign-off 71

1. Executive Summary

Payroll is currently prepared in an Excel worksheet. The payroll officer types each employee's **Total Salary**, then manually adjusts the number of increments in the Basic column until the **Bank Salary** is as close as possible to **80% of Total Salary**. The balance becomes the **Cash Salary**. Employee names, phone numbers, joining dates and designations are maintained in the ERP system and must be integrated with payroll.

The Payroll Management System replaces this worksheet with a controlled application built as **eight independent modules**. Each module owns its own setups and data, reads other modules only through read-only references, and can be maintained, tested and released separately (low coupling, high cohesion). Employee data, including designation, is fetched from the ERP through an API.

Capability	Business outcome	Module
Tier of Employee setup	Payroll classification of employees (for example SM, Senior), each tier linked to a salary scale. The designation itself comes from the ERP and is mapped to a tier.	Module 1
Scale setup	Starting amount and increment amount stored as numbers, with effective dates.	Module 2
Payroll Management	Salary heads (H/R, M/A, Conveyance and others) and payroll parameters are set up here. Employee payroll profile with increment tuning, and payroll run with review, approval and locking.	Module 3
Bonus Management	Bonus types (name with percentage) are set up here. Bonus is processed at any time of the year on gross salary: bank bonus is 80% of the bonus and the rest is cash bonus.	Module 4
Payslip Management	Bank and Cash payslips are generated, printed and stored with each finalized salary run.	Module 5
Salary Certificate Request Approval	Employees request a certificate in the ERP; HR verifies and approves it in Payroll; the status returns to the ERP.	Module 6
Salary Certificate Issuance	Numbered, verifiable certificate; the employee downloads it from the ERP and HR can download it from Payroll. HR can also issue directly.	Module 7
Reports	Payroll, bank and cash sheets, variance, exception, bonus, payslip, certificate, setup history and audit reports.	Module 8

Employee ID, name, phone number, joining date, designation, department and employment status are fetched from the ERP through an API (see Chapter 15, Dependencies); the ERP integration is not a separate module. A summary of what changed from v1.1 is in Document Control.

2. Introduction

2.1 Purpose

This BRD defines the business requirements for Phase 1 of the Payroll Management System so that HR, Finance, Management, the ERP team and the development team share one agreed understanding before design and build begin. For every module it defines the features and their functionality, the consolidated business logic, the process flows, the UI screens, and the user stories mapped to roles and actors.

2.2 Background: How Payroll Works Today

1. The payroll officer enters the **Total Salary** for each employee.
2. A column shows **80% of Total Salary**, the reference for the bank portion.
3. Each employee has a **scale** such as 9000-800 (starting amount 9,000; increment 800).
4. In the **Basic** column the user types the number of increments (for example 9000 + 800 × 2) and adjusts it until Bank Salary comes close to the 80% figure.
5. **H/R** is 50% of Basic, **Conveyance** is 1,000 and **M/A** is 10% of Basic.
6. **Bank Salary** = Basic + H/R + Conveyance + M/A.
7. After Bank Salary is finalized, **Cash Salary** = Total Salary − Bank Salary.
8. A check column (Bank + Cash) is compared with Total and should be zero.

Problems with the current method

- Increment counts are found by trial and error; results vary from 1,400 below to 5,800 above the 80% figure across the sample rows.
- Rates (50%, 10%, 1,000, 80%) are typed into formulas, so a policy change means editing every row.
- No history, no approval and no lock; any cell can be overwritten.
- Employee data is retyped and not linked to the ERP; the sample has duplicate names and mismatched records.
- Row 19 of the sample has a blank reconciliation cell and a check value of 10,000 instead of 0, and the sheet has 19 rows against the 18-employee baseline in the requirements document.
- Payslips, bonus and salary certificates are prepared separately by hand.

2.3 Business Objectives

- Calculate salary components consistently from approved rules and setups.
- Reduce manual effort and error in finding the increment count.
- Use the ERP as the single source of employee identity data, including designation.
- Keep every module independent: its own setups, its own data, minimal and one-directional dependencies.
- Keep a traceable, locked record of every finalized payroll and bonus run.
- Produce Bank and Cash payslips together with each finalized salary run.
- Let employees request salary certificates from the ERP and let HR approve and issue them quickly and with control.

2.4 Scope

In scope (Phase 1)	Out of scope (Phase 1)
Employee data (ID, name, phone, joining date, designation, department, status) fetched from the ERP through an API	Tax and statutory deductions, loans, advances
Tier of Employee setup and Scale setup; ERP designation to tier mapping	Attendance, leave and overtime processing (menu placeholders only, see Section 3.4)
Salary Head setup (H/R, M/A, Conveyance, etc.) and payroll parameters under Payroll Management	Provident fund, gratuity, multi-company
Employee payroll profile, employee-to-scale mapping, Total Salary and increment tuning	Bank transfer file and cash disbursement
Calculation of Basic, heads, Bank Salary, Cash Salary and 80% amount	Accounting / general ledger posting
Payroll run: validate, review, approve, finalize, lock, re-open	Payslip delivery by email or SMS; employee self-service payslip viewing
Bonus Management: Bonus Types (name with percentage), bonus run at any time, bank bonus and cash bonus	Arrears and performance-rating-based bonus rules
Payslips: Bank and Cash payslips generated and printed with the finalized salary	Write-back of payroll data to the ERP (other than certificate request status)
Salary certificate: ERP employee request, HR approval, issuance, download from ERP and Payroll	Mobile application
Reports with Excel / PDF export; worksheet import for migration	Digital signature and public verification portal beyond the QR page

2.5 Stakeholders

Stakeholder	Responsibility
Management / Business Owner	Approves policies, setups, payroll runs and bonus runs.
HR Administrator	Maintains tier, scale and bonus setups; maintains employee tier mapping; verifies and approves salary certificate requests; issues certificates.
Payroll Administrator	Maintains salary head and parameter setups; enters Total Salary, tunes increments, prepares and finalizes payroll and bonus runs; generates and prints payslips.
Finance / Accounts	Reviews totals, bank / cash allocation, bonus split and reconciliation.
ERP / IT Team	Provides the employee data API and the certificate request interface; hosting, security and backup.
Employees	Request salary certificates in the ERP and download approved certificates from the ERP.
System Administrator	Users, roles, audit access and configuration.

2.6 Glossary

Term	Meaning
Total Salary (Gross Salary)	Gross monthly salary entered for the employee; source of the 80% amount, the cash balance and the bonus basis.
Tier of Employee	Payroll classification of an employee (for example SM, Senior) linked to a scale. It is separate from the ERP designation.
Designation	The employee's job title as held in the ERP. It is fetched, never typed in Payroll.

Term	Meaning
Scale	Starting salary amount and increment amount, written as start-increment (for example 9000-800).
Salary Increment	An increase in an employee's salary, entered as a separate Increment Amount and added to the current Total Salary. Not to be confused with the scale increment (N).
Incremented Total Salary	Current Total Salary + Increment Amount; becomes the new Total Salary from the effective date and is the basis for tuning, payroll and bonus.
N (Increment count)	Number of scale increments applied to the starting scale; a whole number, 0 or more.
Basic	Starting scale + (Increment × N).
Salary Head	A salary component other than Basic, set up in Payroll Management (for example H/R, M/A, Conveyance); calculated as a percent of Basic or a fixed amount.
H/R, M/A, Conveyance	House rent allowance (50% of Basic), medical allowance (10% of Basic) and conveyance (fixed 1,000): the current salary heads.
Bank Salary	Basic + all heads flagged as part of Bank Salary; the amount paid through the bank.
Cash Salary	Total Salary − Bank Salary.
80% Reference Amount	Total Salary × Bank target %; the target that Bank Salary should approach.
Variance	Bank Salary − 80% Reference Amount.
Payroll Run	One payroll calculation for a defined period and set of employees.
Rule Snapshot	Copy of setups (scales, heads, parameters) stored with a finalized run.
Bonus Type	A bonus set up as a name with a percentage of gross salary and a bank portion (for example Festival Bonus, 60%).
Bonus Run	One bonus calculation for a Bonus Type, occasion and bonus date; it can be created at any time of the year.
Bank Bonus / Cash Bonus	Bank Bonus = 80% of the bonus; Cash Bonus = the rest, paid in hand cash.
Bank Payslip / Cash Payslip	Payslip showing the bank salary components / payslip showing the cash salary amount.
HR Approval Desk	The Payroll screen where HR verifies and approves salary certificate requests that employees raise in the ERP.
Working Copy	A certificate PDF that HR downloads before approval; marked as not approved and carrying no certificate number.

3. Overall Description

3.1 End-to-End Process Flow

The system runs three independent tracks. The salary payroll cycle runs every period. Bonus can run at any time of the year and only reads the employee's gross salary from the payroll profile. The salary certificate track starts in the ERP when an employee raises a request.

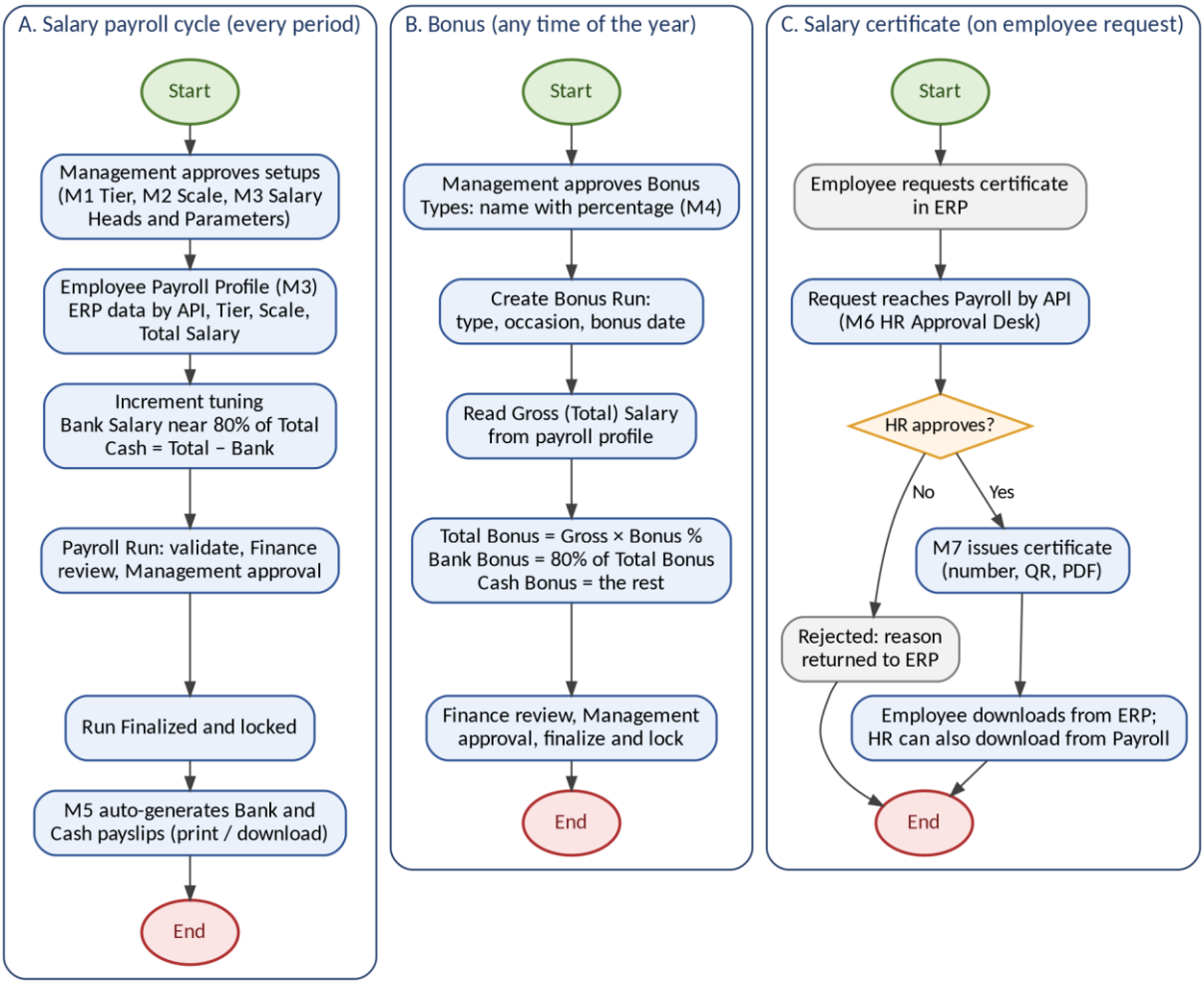


Figure 1: End-to-end process: payroll cycle, bonus and salary certificate

3.2 Modular Architecture and Design Principles

The system is designed so that each module can be built, changed, tested and released without touching the others. The modules follow the principles below.

Principle	How it is applied in this BRD
High cohesion	Every feature of a module serves one business purpose. Salary heads and their rates live in Payroll Management because they exist only to calculate salary; bonus percentages live in Bonus Management because they exist only to calculate bonus; payslip layouts live in Payslip Management; certificate wording lives in Salary Certificate Issuance.
Low coupling	A module never writes another module's data. Cross-module access is read-only and carries only an ID and an effective date. Dependencies are one-directional with no cycles (Figure 2).

Principle	How it is applied in this BRD
Own your setups	Each module owns its setups, so a policy change is made in one place: H/R, M/A and Conveyance in Module 3; bonus name and percentage in Module 4; templates in Modules 5 and 7.
Snapshot, never live link	A module stores a snapshot of what it consumed (scale, head rates, gross salary, bonus %, employee name and designation). Later setup or ERP changes never alter finalized runs, bonus runs, payslips or certificates.
Events, not internal calls	A module announces a state change ("Run Finalized", "Request Approved") and the interested module reacts. Payroll does not call Payslip, so Payslip can be switched off or fail without blocking payroll finalization.
One integration point	Employee data comes from the ERP through one API (Chapter 15). Modules do not keep their own copy of the ERP employee master.

Module responsibility and data ownership

Module	Owns (data and setups)	Reads from	Exposes
M1 Tier of Employee Setup	Tiers; tier-to-scale link; ERP designation to tier mapping	M2 (scale list); ERP (designation list)	Active tiers and their scale
M2 Scale Setup	Scales	Nothing	Active scales
M3 Payroll Management	Salary heads and rates; payroll parameters; employee payroll profile with Total Salary and increment history; payroll runs	M1, M2; ERP (employees)	Gross salary on a date (to M4); finalized run snapshot (to M5, M7); event Run Finalized
M4 Bonus Management	Bonus types (name with percentage); bonus runs	M3 (gross salary only); ERP (employee status)	Finalized bonus runs (to M8)
M5 Payslip Management	Payslip templates; payslips	M3 (finalized run)	Payslip PDFs
M6 Certificate Request Approval	Certificate requests; HR decisions	ERP (requests, employees); M7 (issue and preview service)	Request status and approved certificate to the ERP
M7 Salary Certificate Issuance	Certificate templates; certificates; numbers; verification page	M3 (finalized run); ERP (employee)	Issue / preview service (to M6 and HR direct issue); verification page
M8 Reports	No business data	Read-only on M1 to M7	Reports and exports

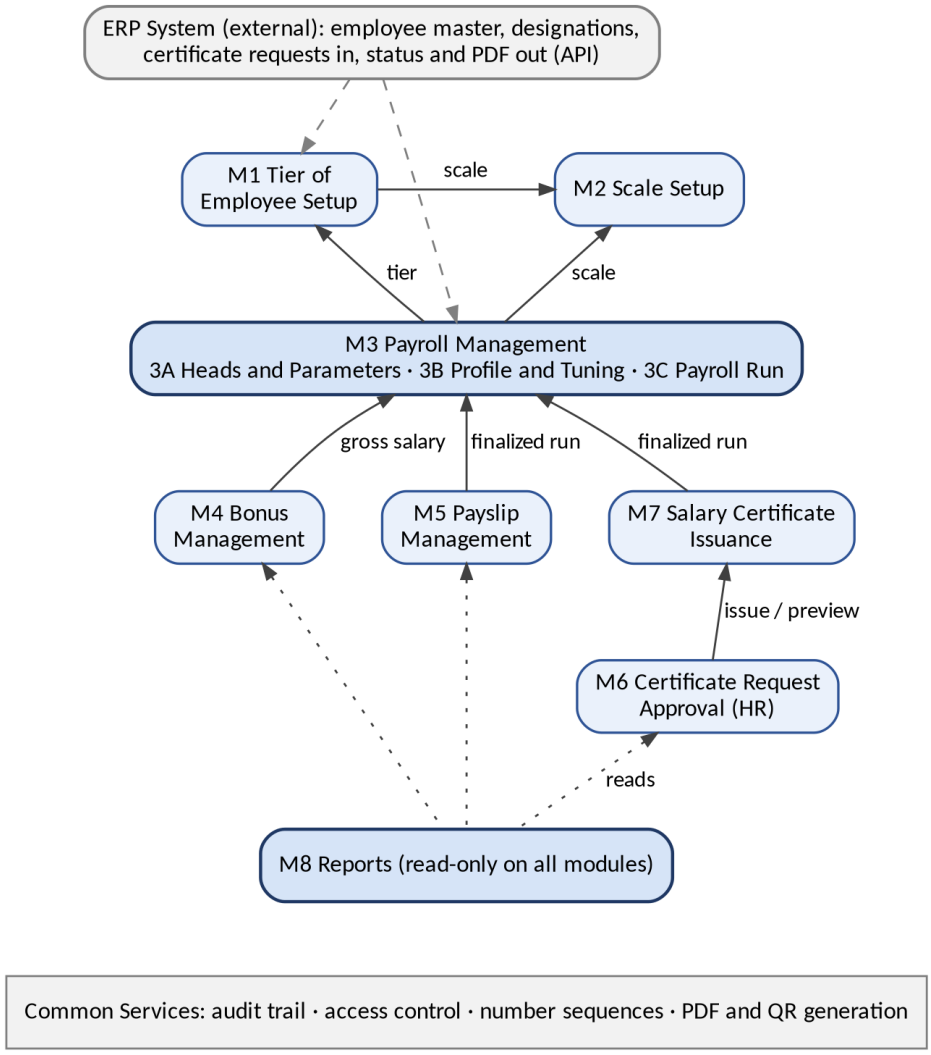


Figure 2: Module architecture and dependencies (an arrow points to the module being read)

3.3 Module Overview

Layer	Module	Features	Summary
Foundation setup	Module 1: Tier of Employee Setup	F1.1 to F1.5	Tiers linked to scales; ERP designation mapping.
Foundation setup	Module 2: Scale Setup	F2.1 to F2.4	Starting amount, increment amount, maximum increments, effective dates.
Core	Module 3: Payroll Management	F3.1 to F3.15	Salary heads and parameters; employee payroll profile and tuning; payroll run.
Transaction	Module 4: Bonus Management	F4.1 to F4.7	Bonus types; bonus run any time; bank bonus and cash bonus.
Output	Module 5: Payslip Management	F5.1 to F5.6	Bank and Cash payslip templates, generation, print, register.
Transaction	Module 6: Salary Certificate Request Approval	F6.1 to F6.6	ERP request intake; HR Approval Desk; status back to ERP.
Output	Module 7: Salary Certificate Issuance	F7.1 to F7.6	Templates, issue, direct issue, register, cancel and re-issue, QR verification.
Report	Module 8: Reports	F8.1 to F8.3	Catalogue of reports, filters and export.

3.4 Application Menu Structure

The menu follows the modules. Items marked Phase 2 are shown as disabled placeholders so that the frontend menu and feature list are ready for them; no Phase 1 logic sits behind them.

Menu group	Menu item	Module	Phase
Setup	Tier of Employee	M1	Phase 1
Setup	Scale	M2	Phase 1
Payroll	Salary Heads	M3	Phase 1
Payroll	Payroll Parameters	M3	Phase 1
Payroll	Employee Payroll Profile	M3	Phase 1
Payroll	Salary Increment	M3	Phase 1
Payroll	Worksheet Import	M3	Phase 1
Payroll	Payroll Run	M3	Phase 1
Bonus	Bonus Types	M4	Phase 1
Bonus	Bonus Run	M4	Phase 1
Payslip	Payslip Templates	M5	Phase 1
Payslip	Generate and Print Payslips	M5	Phase 1
Payslip	Payslip Register	M5	Phase 1
Salary Certificate	Approval Desk (ERP requests)	M6	Phase 1
Salary Certificate	Certificate Templates	M7	Phase 1
Salary Certificate	Issue Certificate (HR direct)	M7	Phase 1
Salary Certificate	Certificate Register	M7	Phase 1
Reports	Report catalogue	M8	Phase 1
Administration	Users and Roles; Audit Log; Configuration	Common Services	Phase 1
Phase 2 placeholder	Attendance	Not built	Phase 2 (disabled)
Phase 2 placeholder	Leave	Not built	Phase 2 (disabled)
Phase 2 placeholder	Overtime	Not built	Phase 2 (disabled)

3.5 Roles, Actors and Permissions

A **role** is a permission group in Payroll. An **actor** is the person or system that performs a step. Several actors can hold the same role, and system actors act without a role.

Actor	Type	Role	Description
HR Officer	Human	HR Administrator	Maintains tier, scale and bonus setups and employee tier mapping; verifies, approves and issues salary certificates.
Payroll Officer	Human	Payroll Administrator	Maintains salary heads and parameters; prepares and finalizes payroll and bonus runs; generates payslips.
Finance Reviewer	Human	Finance / Accounts	Reviews payroll runs and bonus runs; checks bank and cash totals.

Actor	Type	Role	Description
Management Approver	Human	Management	Approves setups, payroll runs, bonus runs and re-open requests.
System Admin	Human	System Administrator	Manages users, roles, configuration and audit access.
Employee (in ERP)	Human, external	ERP Employee (ERP role, no Payroll login)	Raises a salary certificate request in the ERP, tracks it and downloads the approved certificate.
ERP System	System, external	n/a	Master of employee data; sends certificate requests to Payroll; receives status and certificate PDF.
Payroll System	System	n/a	Calculates, validates, numbers, generates PDFs and QR codes, and raises events (Run Finalized, Request Approved).

Permission matrix

V = view, C = create / edit, A = approve, F = finalize, X = export or download. A dash means no access.

Function	HR Admin	Payroll Admin	Finance	Management	Sys Admin	Employee (ERP)
M1 Tier setup; M2 Scale setup	V, C	V	V	V, A	V	-
M3 Salary head and parameter setup	V	V, C	V	V, A	V	-
M3 Employee payroll profile and Total Salary	V	V, C	V	V	-	-
M3 Salary increment: enter / approve	V	V, C	V	V, A	-	-
M3 Payroll run: prepare and tune	-	V, C	V	V	-	-
M3 Payroll run: review	-	V	V, A	V	-	-
M3 Payroll run: approve	-	-	-	V, A	-	-
M3 Payroll run: finalize; request re-open	-	V, F	V	A (re-open)	-	-
M4 Bonus type setup	V, C	V	V	V, A	V	-
M4 Bonus run: prepare	-	V, C	V	V	-	-
M4 Bonus run: review and approve	-	V	V, A (review)	V, A	-	-
M4 Bonus run: finalize	-	V, F	V	V	-	-
M5 Payslip template setup	V, C	V	V	V	-	-
M5 Generate, print and download payslips	V, X	V, C, X	V	V	-	-
M6 Approval Desk: verify, approve, reject	V, A	-	-	V	-	-
M6 Certificate request (raise in ERP, download)	-	-	-	-	-	C, X (in ERP)
M7 Certificate template setup	V, C	-	-	V, A	-	-
M7 Direct issue, register, download, cancel and re-issue	V, C, X	-	-	V	-	-
M8 Reports and export	V, X (own scope)	V, X	V, X	V, X	-	-
Users, roles, audit log, configuration	-	-	-	-	V, C	-

Proposed: segregation of duties
The person who prepares a payroll run or bonus run cannot approve the same run, and an HR user cannot approve a certificate request that he or she raised as an employee (BL-18, BL-23, BL-28). Role names follow the requirements document and can be renamed after sign-off.

3.6 Assumptions and Constraints

- The ERP is the master for employee ID, name, phone, joining date, designation, department and employment status. Payroll never edits them; they are fetched through the ERP API.
- Designations must be clearly defined in the ERP before they are mapped to tiers in Payroll.
- Payroll is processed per period; the period type (monthly or other) is configurable and not assumed.
- Currency is configurable; amounts in this document are shown without currency, as in the worksheet.
- Total Salary (gross) is entered by an authorized user; it is not derived from any other value.
- Calculated fields are read-only; users edit only inputs (tier, Total Salary, increment count, scale override).
- Salary certificate requests originate only in the ERP, on request of the employee.
- Rules marked Proposed are the recommended default until the business signs off.

4. Business Logic (Consolidated)

The business rules of v1.1 are merged into 35 consolidated rules (BL-01 to BL-35), grouped by theme. Each rule states its complete logic in one place, shows which v1.1 rules it merges, and is referenced by the module chapters, the acceptance criteria and the user stories.

4.1 Formula Sheet

Item	Formula	Rule
Salary (Module 3)		
Total Increment	Increment × N	BL-09
Basic	Starting Scale + Total Increment	BL-09
Percent head (for example H/R 50%, M/A 10%)	Basic × head %	BL-08
Fixed head (for example Conveyance 1,000)	Fixed amount from the head setup	BL-08
Incremented Total Salary	Current Total Salary + Increment Amount (entered as a separate input)	BL-10
Bank Salary	Basic + all heads flagged "part of Bank Salary"	BL-11
Cash Salary	Total Salary − Bank Salary	BL-11
80% Reference Amount	Total Salary × Bank target %	BL-12
Variance	Bank Salary − 80% Reference Amount	BL-12
Bonus (Module 4)		
Total Bonus	Gross (Total) Salary × Bonus %	BL-22
Bank Bonus	Total Bonus × Bank portion % (80%)	BL-22
Cash Bonus	Total Bonus − Bank Bonus (the rest, in hand cash)	BL-22

4.2 Business Rules Catalogue

Status key: **Observed** = visible in the worksheet; **Req. doc** = stated in the Requirements Collection; **Review** = stated in the review comments on v1.1 or the clarifications of 27 October 2026; **Proposed** = recommended by Business Analysis and needs approval.

ID	Rule	Merged business logic	Merges (v1.1)	Status
A. Employee data and eligibility (ERP)				
BL-01	ERP employee master	• The ERP is the single master for Employee ID (unique key), name, phone, joining date, designation, department and employment status. Payroll fetches them by API when needed and never edits them. • Phone is shown as 11 digits starting 01 (leading zero added; hyphens and spaces removed). Joining date must be a valid date. • If the ID is not found, appears twice, or the name or joining date is missing, an error is shown and the employee is blocked until the ERP data is corrected. • If the API is unavailable, a message is shown and the user can retry. No payroll data changes; the run, bonus run or certificate waits until the call succeeds.	BR-01, 02, 03, 04, 41	Review, Proposed

ID	Rule	Merged business logic	Merges (v1.1)	Status
BL-02	Eligibility	• Only an employee whom the ERP returns as active and who has a Complete payroll profile can enter a payroll run, bonus run, payslip or certificate. • An employee who becomes inactive stays on historical records but is excluded from new runs, bonus runs and certificates.	BR-05	Proposed
BL-03	Identity snapshot	Finalized runs, bonus runs, payslips and certificates keep the employee name, designation and joining date as they were when the record was created. Later ERP changes do not alter them.	BR-06	Req. doc
B. Tier, scale and effective dating (Modules 1 and 2)				
BL-04	Tier of employee	• Tier code is unique and the name is required. Each tier is linked to one scale at a time through an effective-dated link. • An inactive tier cannot be assigned to employees but stays on past records. Tiers, links and mappings are never deleted, only ended or made inactive. • The designation is never typed in Payroll: it is fetched from the ERP and shown beside the tier. Where an ERP designation is mapped to a tier, that tier is suggested; HR confirms or changes it.	BR-07, 09	Review
BL-05	Scale	• A scale stores the starting amount (greater than 0) and the increment amount (0 or more) as separate numbers; the label (for example 9000-800) is generated. • An optional maximum increment count caps N. Scales are never deleted, only ended or made inactive.	BR-08, 14	Req. doc
BL-06	Employee-to-scale mapping	• The employee payroll profile holds the mapping employee, tier, scale. The employee inherits the scale of the assigned tier. • An authorized user may override the scale for one employee with a mandatory reason; the override is audited and shown on the profile.	BR-09	Review
BL-07	Effective dating and snapshot	• Setups (tier-scale link, scale, head rates, payroll parameters, bonus types, templates) are effective-dated, with no overlapping periods for the same item. • A run, bonus run, payslip or certificate uses the values valid for its period or date and stores them as a snapshot. • Changes to scales, tier links, head rates, parameters and bonus types take effect only after Management approval and never alter finalized or issued records.	BR-10	Proposed
C. Salary composition and tuning (Module 3)				
BL-08	Salary heads	• Salary heads (H/R, M/A, Conveyance and others) are maintained in Payroll Management as setup records, not typed into formulas. Each head has a name, display label, calculation type (percent of Basic or fixed amount), an effective-dated value, and flags: part of Bank Salary, show on payslip, show on certificate. • Current heads: H/R = Basic × 50%; M/A = Basic × 10%; Conveyance = fixed 1,000, the same for every employee. • A new head added by setup is included in the calculation, payslip and certificate according to its flags, without code change.	BR-11, 15, 16, 17	Observed, Review
BL-09	Basic and increment	• Total Increment = Increment × N; Basic = Starting Scale + Total Increment. • N is a whole number, 0 or more, and not above the scale's maximum increments when one is set.	BR-12, 13, 14	Observed
BL-10	Gross (Total) Salary	• Total Salary is entered by an authorized user for each employee, must be greater than 0, and is held as a dated version. It is the gross salary used for the 80% amount, the cash balance and the bonus basis.	BR-19, 31	Observed, Review

ID	Rule	Merged business logic	Merges (v1.1)	Status
		• A salary increment is not typed over the Total Salary. The initial Total Salary stays on record; the Increment Amount (greater than 0) is entered in a separate input, and Incremented Total Salary = current Total Salary + Increment Amount. It is saved as a new dated version after Management approval. • From the effective date the salary procedure runs on the incremented Total Salary: 80% reference, increment tuning, Bank and Cash Salary, payroll run, bonus gross and certificates. A finalized run is never changed; an increment applies from the first payroll period starting on or after its effective date (no pro-rata).		
BL-11	Bank Salary and Cash Salary	• Bank Salary = Basic + all heads flagged part of Bank Salary. After Bank Salary is finalized, Cash Salary = Total Salary − Bank Salary. • Reconciliation: Bank Salary + Cash Salary = Total Salary for every employee and for the run; the check difference must be 0. • Negative Cash Salary is an error and blocks finalization unless an authorized user records an override reason.	BR-18, 22, 23, 24	Observed, Proposed
BL-12	80% reference amount	• Reference Amount = Total Salary × Bank target % (80%, set in Payroll Parameters). Variance = Bank Salary − Reference Amount. • It is informational and never changes Bank or Cash Salary by itself.	BR-11, 20	Observed
BL-13	Increment tuning	• The system calculates the exact N, tests the whole numbers on each side, and suggests the one whose Bank Salary is closest to the Reference Amount (a tie suggests the lower N). • The user accepts the suggestion, picks the other candidate, or types any valid whole number with a mandatory reason. • If N would be below 0, N = 0 and "Basic at scale minimum" is flagged. If the increment is 0, tuning is skipped.	BR-21	Observed, Proposed
BL-14	Rounding and precision	Amounts are held as fixed-point values with two decimals (never floating point). Each component is rounded half-up to the approved unit, and Bank Salary is built from the rounded components. Sample amounts are whole numbers.	BR-25	Proposed
BL-15	Read-only calculations	Calculated fields are read-only and show their formula and component breakdown; users edit only inputs.	BR-26	Req. doc
D. Payroll run control (Module 3)				
BL-16	Run creation	• A run is created for a payroll period. Only one active (not rejected) run can exist per period. • On creation the system selects all eligible employees (BL-02), copies their confirmed Total Salary and N, and stores a rule snapshot of the setups valid for the period.	BR-05, 10, 32	Proposed
BL-17	Validation	• Errors block moving to Reviewed: missing Total Salary, N or scale; negative Cash Salary without override; Bank + Cash not equal to Total; employee inactive in the ERP; duplicate employee in the run. • Warnings do not block: variance above tolerance, Basic at scale minimum, N at maximum.	BR-23, 24	Proposed
BL-18	Status flow and approval	• Draft, Reviewed, Approved, Finalized. Finance reviews, Management approves and the Payroll Administrator finalizes. • The person who prepared a run cannot approve it.	BR-29, 30	Req. doc, Proposed
BL-19	Finalization and correction	• Finalization locks the lines, totals and rule snapshot; history is never overwritten. • A correction uses a controlled re-open (reason, Management approval, new version in Draft, earlier version kept) or a dated adjustment entry for a small correction.	BR-31	Req. doc

ID	Rule	Merged business logic	Merges (v1.1)	Status
		• Finalization announces the event "Run Finalized" so that other modules (payslips) can react.		
E. Bonus (Module 4)				
BL-20	Bonus type setup	• Bonus setup belongs to Bonus Management, not to Payroll. A Bonus Type is a name with a percentage of gross salary (for example Festival Bonus, 60%), plus a bank portion % (default 80%), effective dates and a status. • Changes need Management approval. The percentages are stored with each bonus run, so later changes do not affect it.	BR-11, 27	Review
BL-21	Bonus timing and basis	• A bonus run can be created at any time of the year and any number of times; it does not need a payroll run. • The basis is the gross (Total) Salary of the employee payroll profile effective on the bonus date; the value is stored in the bonus run. • Only ERP-active employees with a Complete payroll profile are included (BL-02).	BR-28	Review, Proposed
BL-22	Bonus calculation	• Total Bonus = Gross Salary × Bonus %. • Bank Bonus = Total Bonus × Bank portion % (80%). Cash Bonus = Total Bonus − Bank Bonus (the rest, paid in hand cash). • Bank Bonus is rounded per BL-14 and Cash Bonus takes the remainder, so Bank Bonus + Cash Bonus always equals Total Bonus.	BR-25, 27	Review
BL-23	Bonus run control	• The bonus run follows the same flow as a payroll run: Draft, Reviewed, Approved, Finalized. The preparer cannot approve it, and finalization locks it. • An employee can be excluded with a reason. A warning appears if the same employee already has an active bonus run for the same Bonus Type and occasion.	BR-29, 30, 31	Proposed
F. Payslip (Module 5)				
BL-24	Payslip generation	• Payslips are generated only from a finalized payroll run, using the stored snapshot, together with the salary generation: automatically when the run is finalized. • Two types are kept: a Bank Payslip for every employee with Bank Salary above 0 and a Cash Payslip for every employee with Cash Salary above 0. • Each payslip has a unique number (BPS/YYYY/NNNNN or CPS/YYYY/NNNNN) and is immutable. • If the Payslip module is unavailable, the run is still finalized; payslips are generated when it is available again.	New (was Phase 2)	Review
BL-25	Payslip content and privacy	• The Bank Payslip shows the bank components (Basic and the heads flagged for payslip) and Bank Salary. The Cash Payslip shows the Cash Salary and an acknowledgement line. Neither shows the other's amount; the template can allow Total Salary on the Bank Payslip (default off). • A payslip contains only its own employee's data and is visible only to authorized roles.	BR-40	Proposed
BL-26	Payslip lifecycle	• Printing and downloading are allowed any number of times; each reprint is logged. • When a run is re-opened and finalized again, earlier payslips become Superseded (kept on file) and new payslips are generated from the new version.	New	Proposed
G. Salary certificate (Modules 6 and 7)				

ID	Rule	Merged business logic	Merges (v1.1)	Status
BL-27	Request from the ERP	• A salary certificate is requested by the employee in the ERP (purpose, addressee) and reaches Payroll through the API with its ERP request ID. A request ID already received is rejected as a duplicate. • Payroll never creates a request on an employee's behalf.	BR-33	Review
BL-28	HR verification and approval	• Every ERP request goes to the HR Approval Desk with status Requested. HR checks that the employee is active in the ERP and has finalized payroll, and reviews the purpose and addressee. • HR approves or rejects; a reason is mandatory for rejection. The approver cannot be the requester. Only approved requests proceed to issuance.	BR-36	Review
BL-29	HR working copy	• HR can download the certificate from Payroll at any time while verifying, without waiting for approval. • A copy downloaded before approval is a working copy, marked "Not approved" and without a certificate number. The employee can download only after approval.	New	Review, Proposed
BL-30	Issuance and availability	• On approval, Module 7 issues the certificate: eligibility check (active employee with finalized payroll), unique number (for example SC/2026/0001, reset yearly), QR verification code and PDF. • Figures come only from the latest finalized payroll; the template decides what prints. Cash Salary and the 80% amount are not printed by default. • Status and PDF are made available to the ERP through the API so that the employee can download the certificate there; HR can download the same certificate from Payroll.	BR-33, 34, 35, 37	Review, Proposed
BL-31	HR direct issue	HR can also issue a certificate directly from Module 7, with no ERP request and no approval, for an eligible employee. The same eligibility, numbering, QR and audit rules apply and the source is recorded as "HR direct".	BR-36	Review
BL-32	Certificate lifecycle	• An issued certificate is immutable; reprinting is allowed and logged. • A change needs Cancel and Re-issue by HR with a mandatory reason; the cancelled number stays on file. • The QR opens a verification page showing the number, employee name, issue date and Valid or Cancelled.	BR-35, 37	Proposed
H. Governance (all modules)				
BL-33	Audit	All creates, edits, status changes, approvals, finalizations, re-opens, overrides, payslip prints and certificate actions are logged with user, date and time, old value and new value. Audit records cannot be edited or deleted.	BR-38	Req. doc
BL-34	Access and privacy	Salary data is visible only to authorized roles on every screen, report and export. Every export is logged, and salary data is not shown in logs or error messages.	BR-40	Req. doc
BL-35	Configuration	Currency, payroll period type, rounding, number formats and head rates are configuration values and are not hard-coded.	BR-39	Req. doc

4.3 Calculation Reference

Salary example

Employee on scale 9000-800, N = 2 (worksheet value), Total Salary 24,000, with the current heads H/R 50%, M/A 10% and Conveyance 1,000:

Step	Formula	Result
Total Increment	800 × 2	1,600
Basic	9,000 + 1,600	10,600
H/R	10,600 × 50%	5,300
Conveyance	Fixed	1,000
M/A	10,600 × 10%	1,060
Bank Salary	10,600 + 5,300 + 1,000 + 1,060	17,960
Cash Salary	24,000 − 17,960	6,040
80% Reference Amount	24,000 × 80%	19,200
Variance	17,960 − 19,200	−1,240

Salary increment example

Employee on scale 9000-800 with current Total Salary 24,000 receives an increment of 2,000 (entered separately). The salary procedure then runs on 26,000:

Step	Formula	Result
Incremented Total Salary	24,000 + 2,000	26,000
80% Reference Amount	26,000 × 80%	20,800
N exact	((20,800 − 1,000) ÷ 1.6 − 9,000) ÷ 800	4.22
Suggested N (4: variance −280; 5: variance +1,000)	Smallest absolute variance	4
Basic; Bank Salary	9,000 + 800 × 4; 12,200 + 6,100 + 1,000 + 1,220	12,200; 20,520
Cash Salary	26,000 − 20,520	5,480

Bonus example

Bonus on gross salary with an illustrative Bonus Type of 60% and bank portion 80%. The cash bonus is the remainder, so the two parts always add up to the total bonus.

Case	Gross (Total) Salary	Total Bonus (60%)	Bank Bonus (80%)	Cash Bonus (rest)
Example 1	24,000	14,400	11,520	2,880
Example 2	70,000	42,000	33,600	8,400
Example 3 (rounding)	10,001	6,001 (6,000.6 rounded)	4,801 (4,800.8 rounded)	1,200 (6,001 − 4,801)
Baseline, 18 employees (derived)	**693,500**	**416,100**	**332,880**	**83,220**

> **Note**
> Bonus values are derived for testing and are not a policy approval. The 60% and 80% are illustrative until the Bonus Types are confirmed (Chapter 19).

5. Module 1: Tier of Employee Setup

5.1 Purpose

Maintain the payroll classification **Tier of Employee** (for example SM, Senior) and link each tier to a salary scale. This module replaces the Designation Setup of v1.1. The employee's designation is not maintained here: it is fetched from the ERP and mapped to a tier, and the tier decides the scale the employee inherits in the payroll profile.

5.2 Features

ID	Feature	Functionality	Primary actor
F1.1	Tier master	Create and edit a tier with code, name and description. Code is unique; name is required.	HR Officer
F1.2	Tier-to-scale link	Link each tier to one scale from Module 2 with an effective-from date. Earlier links stay as history. A new link applies only from its effective date and takes effect after Management approval.	HR Officer, Management Approver
F1.3	ERP designation mapping (optional)	Fetch the designation list from the ERP and map one or more designations to a tier. In the employee payroll profile, the mapped tier is suggested; HR confirms or changes it. Designations not yet mapped are listed for attention.	HR Officer
F1.4	Tier status	Make a tier Active or Inactive. An inactive tier cannot be assigned to employees but stays on past records. Tiers are never deleted.	HR Officer
F1.5	Search and history	Search by code, name, scale or designation; view the link and mapping history of a tier.	HR Officer

5.3 Business Logic

- Tier code is unique; each tier is linked to one scale at a time through an effective-dated link (BL-04, BL-07).
- The employee-to-scale mapping is held in the payroll profile (Module 3): employee, tier, scale. A scale override for one employee needs a reason (BL-06).
- The designation comes from the ERP only. Mapping goes designation, tier, scale, so Scale Setup stays independent of the ERP (BL-04).
- This module reads only the scale list from Module 2 and the designation list from the ERP. It does not read employee or payroll data, so it has no dependency on Module 3.
- Every create, edit and status change is logged (BL-33).

5.4 UI Screen Specification

Screen 1.1: Tier List and Form

Role: HR Administrator | Menu: Setup > Tier of Employee

Field / Control	Type	Options / Format / Behaviour	Mandatory
Tier Code	Text (20)	Unique; letters, digits, hyphen.	Yes
Tier Name	Text (100)	Unique per organization, for example SM, Senior.	Yes
Description	Text area (250)	Free text.	No
Status	Toggle	Active / Inactive; default Active.	Yes

Field / Control	Type	Options / Format / Behaviour	Mandatory
Current Scale	Read-only text	Scale label (for example 9000-800) valid today.	N/A
Search / Filter	Text, dropdown	By code, name, scale, status.	No
Save / Cancel	Buttons	Save logs the change.	N/A

Screen 1.2: Tier-to-Scale Link

Role: HR Administrator (Management approves) | Opens from Screen 1.1

Field / Control	Type	Options / Format / Behaviour	Mandatory
Tier	Read-only	Selected tier.	N/A
Scale	Dropdown	Active scales from Module 2; shows the label.	Yes
Effective From	Date	Cannot overlap an existing link of the same tier.	Yes
Approval Status	Badge	Draft / Approved; only Approved links are used.	N/A
Link history	Grid	All earlier links with dates and approver.	N/A

Screen 1.3: ERP Designation Mapping

Role: HR Administrator | Opens from Screen 1.1

Field / Control	Type	Options / Format / Behaviour	Mandatory
ERP Designation	Multi-select	List fetched from the ERP API; read-only names.	Yes
Mapped Tier	Read-only	Tier currently mapped; a designation maps to one active tier only.	N/A
Unmapped designations	List	Designations returned by the ERP with no tier.	N/A
ERP unavailable	Message	Shown with Retry; existing mappings stay usable.	N/A

5.5 User Stories

ID	User story	Role	Actor	Feature	Priority
US-1.1	As an HR Administrator, I want to create and maintain employee tiers (for example SM, Senior) so that payroll classifies employees independently of ERP designations.	HR Administrator	HR Officer	F1.1	Must
US-1.2	As an HR Administrator, I want to link a tier to a scale with an effective date so that employees of that tier inherit the correct scale.	HR Administrator	HR Officer	F1.2	Must
US-1.3	As Management, I want to approve tier-to-scale links before they apply so that salary structure changes are controlled.	Management	Management Approver	F1.2	Must
US-1.4	As an HR Administrator, I want to map ERP designations to a tier so that the tier is suggested when I set up an employee payroll profile.	HR Administrator	HR Officer	F1.3	Should
US-1.5	As the Payroll System, I need to fetch the designation list from the ERP so that no designation is ever typed in Payroll.	System (automated)	Payroll System	F1.3	Must
US-1.6	As an HR Administrator, I want to inactivate a tier without losing history so that past payroll records stay intact.	HR Administrator	HR Officer	F1.4	Should

6. Module 2: Scale Setup

6.1 Purpose

Hold each salary scale as two numbers, a starting amount and an increment amount, replacing the typed labels such as 9000-800 in the worksheet. This module has no dependency on any other module.

6.2 Features

ID	Feature	Functionality	Primary actor
F2.1	Scale master	Create and edit a scale: code, starting amount, increment amount and optional maximum increments. The label start-increment is generated.	HR Officer
F2.2	Effective dating and history	Each scale has effective-from and optional effective-to dates; periods of the same code cannot overlap; changes need Management approval.	HR Officer, Management Approver
F2.3	Scale status	Make a scale Active or Inactive. Scales are never deleted; an inactive scale cannot be newly linked or assigned.	HR Officer
F2.4	Opening scales	Create the scales found in the worksheet at migration: 9000-800, 10000-900, 12000-1000, 15000-1100.	HR Officer

6.3 Business Logic

- Starting amount and increment amount are separate numeric fields; the label is generated (BL-05).
- An optional Maximum Increments value caps N in tuning (BL-05, BL-09).
- Scales are effective-dated; periods for the same scale code cannot overlap; approval is required (BL-07).
- Where a scale is used is shown in the Setup Usage Report (Module 8), so this module does not need to read other modules.

6.4 UI Screen Specification

Screen 2.1: Scale List and Form

Role: HR Administrator (Management approves) | Menu: Setup > Scale

Field / Control	Type	Options / Format / Behaviour	Mandatory
Scale Code	Text (20)	Unique; for example S-9000-800.	Yes
Starting Amount	Decimal (12,2)	Greater than 0.	Yes
Increment Amount	Decimal (12,2)	0 or more. If 0, tuning is skipped.	Yes
Scale Label	Read-only text	Generated: start-increment.	N/A
Maximum Increments	Whole number	Optional cap for N.	No
Effective From / To	Date	No overlap for the same code; To is optional.	Yes / No
Approval Status	Badge	Draft / Approved.	N/A
Status	Toggle	Active / Inactive.	Yes

6.5 User Stories

ID	User story	Role	Actor	Feature	Priority
US-2.1	As an HR Administrator, I want to define a scale as a starting amount and an increment amount so that salary scales are numbers, not typed labels.	HR Administrator	HR Officer	F2.1	Must
US-2.2	As an HR Administrator, I want to set a maximum number of increments so that tuning never goes beyond the scale.	HR Administrator	HR Officer	F2.1	Should
US-2.3	As an HR Administrator, I want to revise a scale from a future date so that old payroll keeps using the old values.	HR Administrator	HR Officer	F2.2	Must
US-2.4	As Management, I want to approve scale changes before they apply so that amounts are controlled.	Management	Management Approver	F2.2	Must
US-2.5	As an HR Administrator, I want to inactivate a scale so that it cannot be used for new assignments while history stays intact.	HR Administrator	HR Officer	F2.3	Should

7. Module 3: Payroll Management

7.1 Purpose

Everything that calculates salary lives in this module, in three cohesive parts: **3A Salary Structure Setup** (salary heads such as H/R, M/A and Conveyance, and payroll parameters), **3B Employee Payroll Profile and Increment Tuning**, and **3C Payroll Run**. The percentage setup for H/R, M/A, Conveyance and similar heads belongs here and nowhere else. Bonus percentages are not part of this module; they belong to Module 4.

Dependencies: Module 1 (tier and its scale), Module 2 (scale amounts) and the ERP (employee data, Chapter 15). Other modules read from this module only through its published outputs: gross salary on a date (Module 4), the finalized run snapshot (Modules 5 and 7) and the event Run Finalized.

7.2 Features

3A Salary Structure Setup

ID	Feature	Functionality	Primary actor
F3.1	Salary Head Setup	Add and maintain salary heads such as H/R, M/A and Conveyance. Each head has a code, name, display label, calculation type (percent of Basic or fixed amount), flags (part of Bank Salary, show on payslip, show on certificate), display order and status. H/R, M/A and Conveyance are created at migration; further heads can be added without code change.	Payroll Officer
F3.2	Head rate versions	Each head holds effective-dated values (for example H/R 50%, M/A 10%, Conveyance 1,000). Periods cannot overlap. A new value applies from its date after Management approval.	Payroll Officer, Management Approver
F3.3	Payroll Parameter Setup	Maintain Bank target % (80%), rounding method and unit, currency, payroll period type and the variance tolerance for warnings. Effective-dated and approved by Management.	Payroll Officer, Management Approver

3B Employee Payroll Profile and Increment Tuning

ID	Feature	Functionality	Primary actor
F3.4	Employee selection from the ERP	Search an employee in the ERP through the API; display ID, name, phone, joining date, designation and status read-only. Employees with invalid or duplicate ERP data, or inactive in the ERP, are blocked.	Payroll Officer
F3.5	Tier and scale mapping	Assign the employee a tier (suggested from the ERP designation mapping). The scale is inherited from the tier; an authorized user can override it with a reason. This is the employee-to-scale mapping.	Payroll Officer
F3.6	Total (gross) Salary entry	Enter Total Salary for the employee with an effective-from date; each change is a new version and the history is kept.	Payroll Officer
F3.7	Increment tuning	The system suggests the increment count N so that Bank Salary is closest to the 80% Reference Amount, shows the candidates, and lets the user accept, choose or override with a reason.	Payroll Officer
F3.8	Salary calculation preview	Show Basic, every head, Bank Salary, Cash Salary, 80% amount and variance with the formula for each, plus warnings (negative cash, Basic at scale minimum, N at maximum, variance above tolerance).	Payroll Officer

ID	Feature	Functionality	Primary actor
F3.15	Salary increment	Record an increment as a separate input. The screen shows the current Total Salary; the user enters the Increment Amount and effective date; Incremented Total Salary = current Total Salary + Increment Amount (read-only). After Management approval it becomes a new profile version. The initial salary and every increment stay in the history. From the effective date, tuning, payroll run, bonus gross and certificates use the incremented Total Salary.	Payroll Officer, Management Approver
F3.9	Worksheet import	Import Total Salary and N from the Excel worksheet for migration, mapping each row to an ERP employee ID by confirmation; the system recalculates every component.	Payroll Officer

3C Payroll Run

ID	Feature	Functionality	Primary actor
F3.10	Run creation and calculation	Create a run for a period; select eligible employees; store the rule snapshot; calculate all lines.	Payroll Officer
F3.11	Validation and exceptions	Validate every line; errors block review and warnings do not; list exceptions per employee.	Payroll Officer
F3.12	Review, approval and finalization	Draft, Reviewed (Finance), Approved (Management), Finalized (Payroll Administrator). Finalization locks the run and raises the event Run Finalized.	Finance Reviewer, Management Approver, Payroll Officer
F3.13	Re-open and adjustment	Correct a finalized run by a re-open (reason, Management approval, new version) or a dated adjustment entry. Earlier versions are kept.	Payroll Officer, Management Approver
F3.14	Run totals and reconciliation	Totals row (scales, increments, Basic, heads, Bank, Cash, Total, 80% amount, headcount) and the check Bank + Cash = Total per line and for the run.	Payroll Officer, Finance Reviewer

7.3 Business Logic

Salary composition

Basic = Starting Scale + (Increment × N)

Head = Basic × head % (percent heads) or the fixed amount (fixed heads)

Bank Salary = Basic + all heads flagged "part of Bank Salary"

Cash Salary = Total Salary − Bank Salary **Check:** Bank + Cash − Total = 0

- Heads, rates and parameters are read from the setups valid for the period and stored as a rule snapshot with the run (BL-07, BL-08, BL-12).
- The employee inherits the scale of the assigned tier; an override needs a reason (BL-06). Total Salary must be greater than 0 (BL-10).
- A salary increment is entered as a separate Increment Amount and added to the current Total Salary; the new Total Salary is a new dated version and the whole salary procedure runs on it from the effective date (BL-10).
- Calculated fields are read-only and show their formula (BL-15). Amounts are rounded half-up to the approved unit (BL-14).
- Negative Cash Salary blocks finalization unless an authorized override with a reason is recorded (BL-11).

- Only one active run per period (BL-16). Errors block review and warnings do not (BL-17). The preparer cannot approve (BL-18).
- A finalized run is locked; corrections use re-open or an adjustment entry (BL-19).

Increment tuning algorithm

1. Ref = Total Salary × Bank target %.
2. BasicTarget = (Ref − sum of fixed heads) ÷ (1 + sum of percent heads). With the current heads this is (Ref − 1,000) ÷ 1.6.
3. N_exact = (BasicTarget − Starting Scale) ÷ Increment.
4. Candidates: N_low = max(0, floor(N_exact)) and N_high = max(0, ceiling(N_exact)), both capped at Maximum Increments.
5. For each candidate calculate Basic, heads, Bank, Cash and Variance = Bank − Ref.
6. Suggest the candidate with the smallest absolute variance; if equal, suggest the lower N.
7. If N_exact is below 0, use N = 0 and flag "Basic at scale minimum". If Increment is 0, skip tuning and show the variance.

The algorithm is shown as a flow in Figure 4 (Section 7.4).

Worked examples

Scale	Total	Ref (80%)	N exact	Suggested N	Basic	H/R	M/A	Bank	Cash	Variance
9000-800	24,000	19,200	2.97	3	11,400	5,700	1,140	19,240	4,760	+40
10000-900	70,000	56,000	27.08	27	34,300	17,150	3,430	55,880	14,120	−120
9000-800	20,000	16,000	0.47	0	9,000	4,500	900	15,400	4,600	−600

Decision needed (Chapter 19, item 1)
The worksheet shows Bank Salary both below and above the 80% amount. For 9000-800 with Total 24,000 the worksheet used N = 2 (Bank 17,960, variance −1,240) while N = 3 gives 19,240 (variance +40). The system suggests the nearest count and keeps the worksheet value available as an override until the business states whether Bank Salary may exceed the 80% amount.

7.4 Process Flow

Figure 3 shows how an employee payroll profile is completed, Figure 4 the increment tuning, and Figure 5 the payroll run from creation to finalization and later correction.

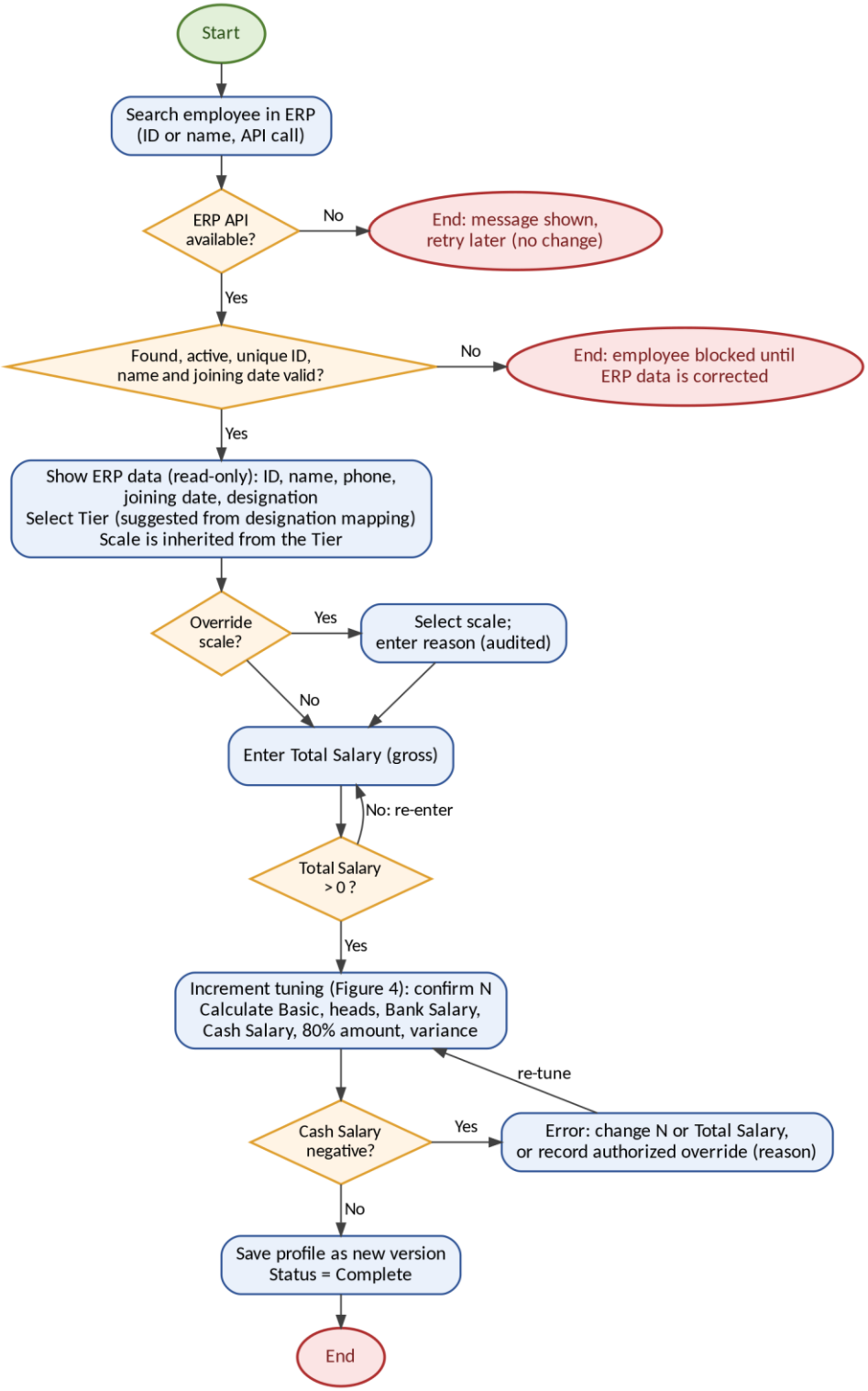


Figure 3: Employee payroll profile flow (ERP fetch, tier, scale, Total Salary, tuning)

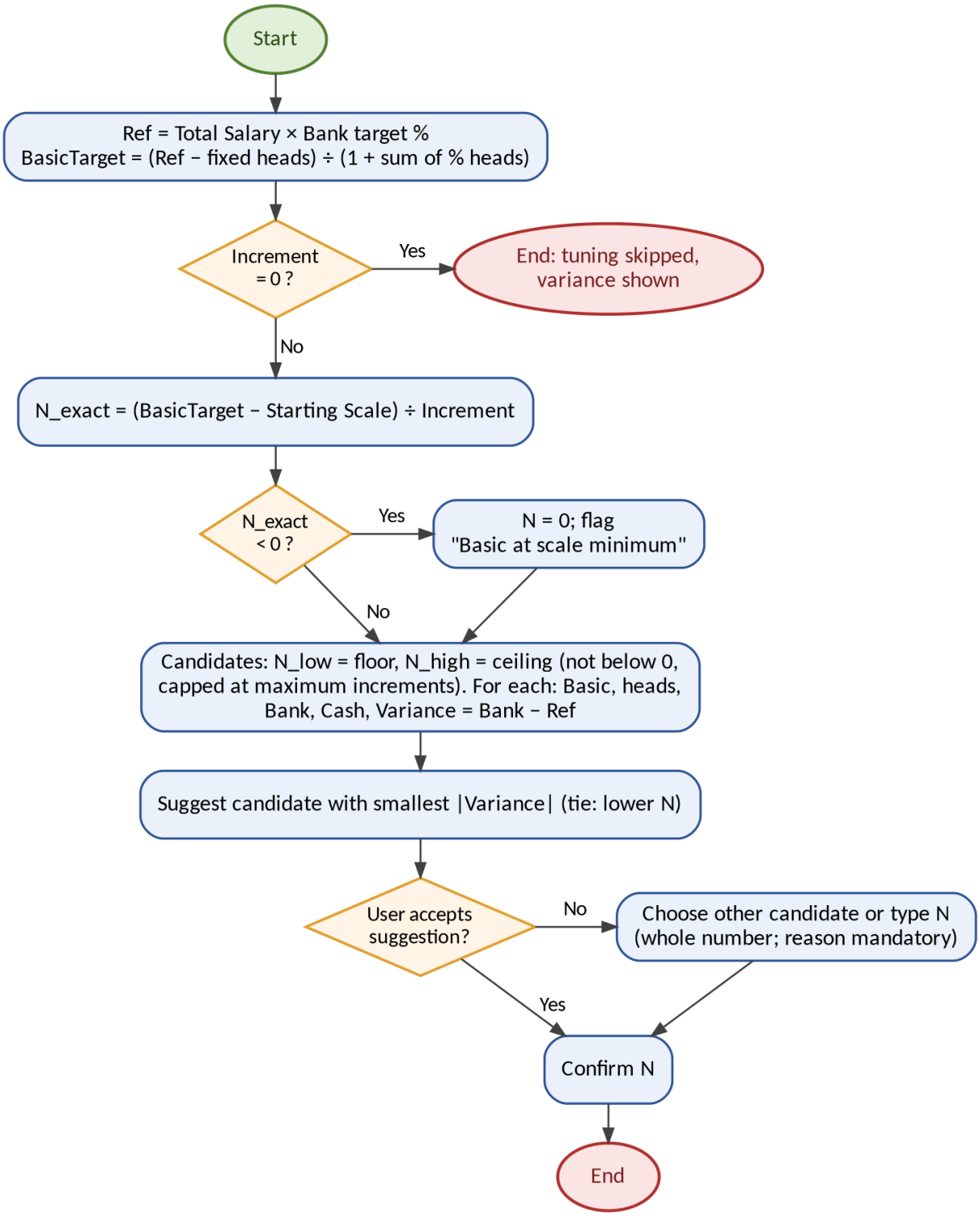


Figure 4: Increment tuning logic

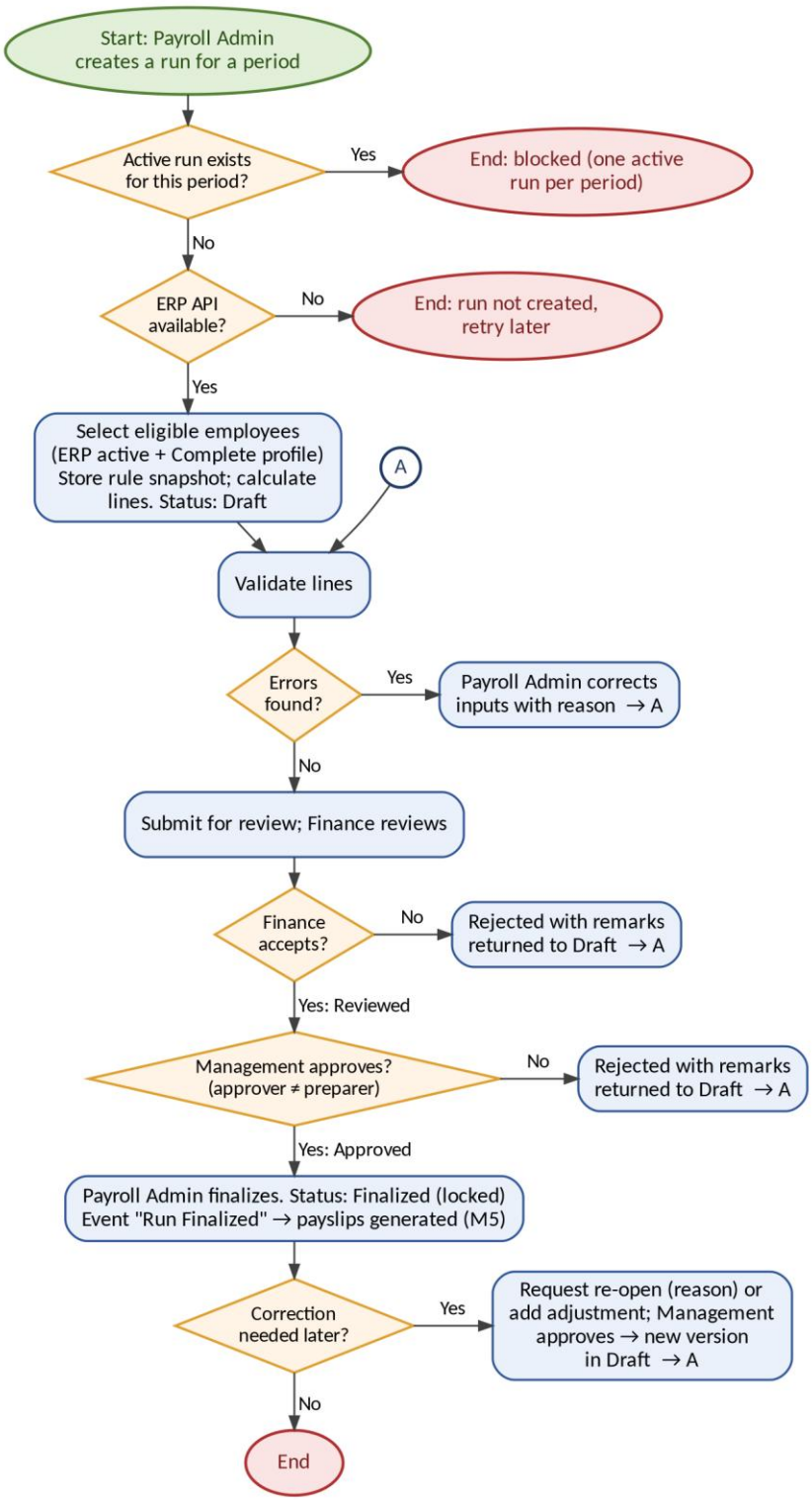


Figure 5: Payroll run flow with status, conditions and re-open

7.5 UI Screen Specification

Screen 3.1: Salary Head List and Form

Role: Payroll Administrator (Management approves rates) | Menu: Payroll > Salary Heads

Field / Control	Type	Options / Format / Behaviour	Mandatory
Head Code	Text (20)	Unique, for example HR, MA, CONV.	Yes

Field / Control	Type	Options / Format / Behaviour	Mandatory
Head Name	Text (60)	For example House Rent, Medical Allowance, Conveyance.	Yes
Display Label	Text (30)	Label printed on payslip and certificate, for example H/R, M/A.	Yes
Calculation Type	Dropdown	Percent of Basic / Fixed amount. Locked once the head is used in a finalized run.	Yes
Part of Bank Salary	Checkbox	Default on. Cash Salary = Total − Bank Salary.	Yes
Show on Payslip / Certificate	Checkboxes	Controls the head on the Bank Payslip and on the certificate figures.	No
Display Order	Whole number	Order of calculation and printing.	Yes
Status	Toggle	Active / Inactive. Heads are never deleted.	Yes
Rate Versions	Grid	Value (percent 0 to 100 or fixed amount 0 or more), Effective From / To (no overlap), Approval Status.	Yes

Screen 3.2: Payroll Parameter Setup

Role: Payroll Administrator (Management approves) | Menu: Payroll > Payroll Parameters

Field / Control	Type	Options / Format / Behaviour	Mandatory
Bank target %	Decimal	0 to 100; current 80.	Yes
Rounding Method / Unit	Dropdown	Half-up to unit (proposed); unit 1, 0.5 or 0.01.	Yes
Currency	Text + code	Configured by business; not assumed.	Yes
Payroll Period Type	Dropdown	Monthly or as approved.	Yes
Variance Tolerance %	Decimal	Warning threshold on the 80% amount; proposed 5.	Yes
Effective From / To	Date	No overlap per parameter.	Yes / No
Approval Status	Badge	Draft / Approved.	N/A

Screen 3.3: Employee Payroll Profile List

Role: Payroll Administrator | Menu: Payroll > Employee Payroll Profile

Field / Control	Type	Options / Format / Behaviour	Mandatory
Employee (search by ID or name)	Searchable dropdown	Data from the ERP API; shows Employee ID and name.	Yes
ERP Designation	Read-only	Fetched from the ERP.	N/A
Tier	Read-only	Assigned tier.	N/A
Scale	Read-only	Inherited or overridden scale label.	N/A
Total Salary	Read-only	Current version.	N/A
Profile Status	Badge	Pending / Complete / Inactive.	N/A
Filter	Dropdown	Profile status, tier, scale, variance flag.	No

Screen 3.4: Profile and Tuning Panel

Role: Payroll Administrator | Opens from Screen 3.3

Field / Control	Type	Options / Format / Behaviour	Mandatory
Employee ID, Name, Phone, Joining Date, Designation	Read-only	Fetched from the ERP API.	N/A
Tier	Dropdown	Active tiers (Module 1); the tier mapped from the ERP designation is pre-selected.	Yes
Scale / Scale Override Reason	Read-only / Dropdown + Text (250)	Scale inherited from the tier; override needs permission and a reason.	Yes / Conditional
Total Salary	Decimal (12,2)	Greater than 0. Editing refreshes tuning.	Yes
Effective From	Date	Date from which this version applies.	Yes
80% Reference Amount, N Exact	Read-only decimal	Total × Bank target %; for information.	N/A
Candidate Table	Read-only grid	N_low and N_high with Basic, Bank, Cash, Variance; suggested row highlighted.	N/A
Increment Count (N)	Whole number	Select a candidate or type a value; not above the maximum.	Yes
Override Reason	Text (250)	Mandatory when N differs from the suggestion.	Conditional
Basic, each head, Bank Salary, Cash Salary, Variance	Read-only decimal	Formula shown on hover or expand; Cash shown red if negative.	N/A
Warnings	Message list	Negative cash, Basic at scale minimum, N at maximum, variance above tolerance.	N/A
Save / Cancel	Buttons	Save creates a new profile version.	N/A

Screen 3.9: Salary Increment

Role: Payroll Administrator (Management approves) | Opens from Screen 3.3 or Screen 3.4

Field / Control	Type	Options / Format / Behaviour	Mandatory
Employee	Read-only	ID, name, ERP designation, tier, scale.	N/A
Current Total Salary	Read-only decimal	Version effective on the chosen date.	N/A
Increment Amount	Decimal (12,2)	Separate input; greater than 0.	Yes
Reason	Text (250)	For example annual increment, promotion.	No
Incremented Total Salary	Read-only decimal	Current Total Salary + Increment Amount.	N/A
Effective From	Date	Applies from the first payroll period starting on or after this date; cannot change a finalized run.	Yes
Re-tune Preview	Read-only grid	Suggested N, Basic, Bank, Cash and variance on the incremented Total Salary; N is confirmed in Screen 3.4.	N/A
Approval Status	Badge	Draft / Approved; only Approved increments create a new version.	N/A
Salary History	Grid	Initial Total Salary, each Increment Amount, resulting Total Salary, effective date, approver.	N/A

Field / Control	Type	Options / Format / Behaviour	Mandatory
Save / Submit for Approval / Cancel	Buttons	Logged in the audit trail.	N/A

Screen 3.5: Worksheet Import

Role: Payroll Administrator | Menu: Payroll > Worksheet Import

Field / Control	Type	Options / Format / Behaviour	Mandatory
File	File upload	Excel worksheet (xlsx).	Yes
Row mapping	Grid	Each row mapped to an ERP employee ID by confirmation (names repeat).	Yes
Imported fields	Read-only	Total Salary and N; all other values recalculated.	N/A
Issues	List	Missing mapping, blank reconciliation, rows above the 80% amount (override reason required).	N/A
Import / Cancel	Buttons	Import creates profile versions.	N/A

Screen 3.6: Payroll Run List and Create

Role: Payroll Administrator | Menu: Payroll > Payroll Run

Field / Control	Type	Options / Format / Behaviour	Mandatory
Run No	Auto text	PR-YYYY-NNN.	N/A
Payroll Period	Period picker	Per Payroll Parameters period type; one active run per period.	Yes
Run Name	Text (60)	For example the month and year.	No
Headcount	Read-only	Included employees.	N/A
Status	Badge	Draft, Reviewed, Approved, Finalized, Rejected.	N/A
Version	Read-only	Increases on re-open.	N/A
Create Run	Button	Builds lines from Complete profiles of active ERP employees.	N/A

Screen 3.7: Payroll Run Detail

Role: Payroll Administrator, Finance, Management (by status) | Opens from Screen 3.6

Field / Control	Type	Options / Format / Behaviour	Mandatory
Sl, Employee ID, Name, Designation, Tier	Read-only	Designation fetched from the ERP and stored with the line.	N/A
Scale, Increment, N	Read-only	From the confirmed profile; editable only in Draft by Payroll Administrator, with reason.	N/A
Total Salary	Decimal	Editable in Draft only, with reason; greater than 0.	Yes
Total Increment, Basic, each head	Read-only decimal	BL-08, BL-09.	N/A
Bank Salary / Cash Salary	Read-only decimal	BL-11; Cash red if negative.	N/A
80% Amount, Variance	Read-only decimal	BL-12.	N/A
Check (Bank + Cash − Total)	Read-only decimal	Must be 0.	N/A

Field / Control	Type	Options / Format / Behaviour	Mandatory
Flags	Icon list	Error or warning with message.	N/A
Totals Row	Read-only	Per column sums and headcount.	N/A
Validate / Submit for Review / Review / Approve / Reject / Finalize	Buttons	Shown by status and role; Reject needs remarks.	N/A
Request Re-open / Add Adjustment	Buttons	Finalized runs only; reason required; Management approves a re-open.	Reason: Yes
Print Payslips	Link	Finalized runs only; opens Module 5 (navigation only).	N/A
Export	Button	Excel or PDF register.	N/A

Screen 3.8: Approval History

Role: all run roles (view) | Opens from Screen 3.7

Field / Control	Type	Options / Format / Behaviour	Mandatory
Action	Read-only	Created, Submitted, Reviewed, Approved, Rejected, Finalized, Re-opened.	N/A
User, Date and Time, Remarks	Read-only	From the audit log.	N/A

7.6 Validations and Exceptions

Condition	Type	System behaviour
Increment Amount zero, negative or missing	Error	Cannot save; a reduction or correction is made as a new Total Salary version with a reason.
Increment effective date inside a finalized period	Error	Not applied to that run; use the next period or request a re-open.
Total Salary missing or zero	Error	Profile cannot be saved; in a run the employee is not included and is listed in the exception report.
N not a whole number, negative or above the maximum	Error	Cannot save.
Tier without a scale and no override	Error	Cannot save.
Employee invalid, duplicate or inactive in the ERP	Error	Employee blocked or removed from the run after confirmation.
ERP API unavailable	Error	Profile or run cannot be created; user retries; nothing changes.
Cash Salary below 0	Error	Blocks Review; authorized override with reason allowed.
Bank + Cash does not equal Total	Error	Blocks Review; calculation fault to be investigated.
Second run for the same period	Error	Not allowed while another run is active.
Preparer tries to approve own run	Error	Action not available.
Variance above tolerance	Warning	Shown to the reviewer; does not block.
Basic at scale minimum or N at maximum	Warning	Informational.

7.7 User Stories

ID	User story	Role	Actor	Feature	Priority
US-3.1	As a Payroll Administrator, I want to add and maintain salary heads (H/R, M/A, Conveyance and others) so that salary components are configured, not hard-coded.	Payroll Administrator	Payroll Officer	F3.1	Must
US-3.2	As a Payroll Administrator, I want each head to have effective-dated rates so that a rate change applies only from its date.	Payroll Administrator	Payroll Officer	F3.2	Must
US-3.3	As Management, I want to approve head rates and payroll parameters before they apply so that salary policy is controlled.	Management	Management Approver	F3.2, F3.3	Must
US-3.4	As a Payroll Administrator, I want to set the bank target %, rounding, currency, period type and variance tolerance so that calculation rules are configurable.	Payroll Administrator	Payroll Officer	F3.3	Must
US-3.5	As a Payroll Administrator, I want to search an employee in the ERP and see the ERP data, including designation, so that I never retype employee data.	Payroll Administrator	Payroll Officer	F3.4	Must
US-3.6	As the Payroll System, I need to block employees with invalid, duplicate or inactive ERP data so that bad data never enters payroll.	System (automated)	Payroll System	F3.4	Must
US-3.7	As a Payroll Administrator, I want to assign a tier (suggested from the ERP designation) and inherit its scale so that the employee-to-scale mapping is consistent.	Payroll Administrator	Payroll Officer	F3.5	Must
US-3.8	As a Payroll Administrator, I want to override an employee's scale with a reason so that exceptions are possible but traceable.	Payroll Administrator	Payroll Officer	F3.5	Should
US-3.9	As a Payroll Administrator, I want to enter the Total Salary with an effective date so that salary changes keep their history.	Payroll Administrator	Payroll Officer	F3.6	Must
US-3.21	As a Payroll Administrator, I want to enter a salary increment as a separate amount added to the current Total Salary so that the initial salary and the increment are both on record.	Payroll Administrator	Payroll Officer	F3.15	Must
US-3.22	As Management, I want to approve a salary increment before it applies so that salary changes are controlled.	Management	Management Approver	F3.15	Must
US-3.23	As the Payroll System, I need to run tuning, payroll and bonus on the incremented Total Salary from its effective date so that the salary procedure uses the new salary.	System (automated)	Payroll System	F3.15	Must
US-3.24	As an HR Administrator, I want to see an employee's salary history (initial salary and increments) so that I can answer salary queries.	HR Administrator	HR Officer	F3.15	Should
US-3.10	As a Payroll Administrator, I want the system to suggest the increment count closest to 80% so that I stop adjusting by trial and error.	Payroll Administrator	Payroll Officer	F3.7	Must
US-3.11	As a Payroll Administrator, I want to override the suggested N with a reason so that real-world exceptions are recorded.	Payroll Administrator	Payroll Officer	F3.7	Must
US-3.12	As Finance, I want to see every component with its formula and warnings so that I can verify the calculation.	Finance / Accounts	Finance Reviewer	F3.8	Should

ID	User story	Role	Actor	Feature	Priority
US-3.13	As a Payroll Administrator, I want to import Total Salary and N from the Excel worksheet so that migration is quick and checked.	Payroll Administrator	Payroll Officer	F3.9	Should
US-3.14	As a Payroll Administrator, I want to create a run for a period that picks all eligible employees so that I do not select them by hand.	Payroll Administrator	Payroll Officer	F3.10	Must
US-3.15	As a Payroll Administrator, I want errors and warnings listed per employee so that I can fix problems before review.	Payroll Administrator	Payroll Officer	F3.11	Must
US-3.16	As Finance, I want to review the run and reject it with remarks so that errors are caught before approval.	Finance / Accounts	Finance Reviewer	F3.12	Must
US-3.17	As Management, I want to approve a run that I did not prepare so that duties are separated.	Management	Management Approver	F3.12	Must
US-3.18	As a Payroll Administrator, I want to finalize and lock an approved run so that history cannot be overwritten.	Payroll Administrator	Payroll Officer	F3.12	Must
US-3.19	As a Payroll Administrator, I want to request a re-open or add an adjustment with a reason so that corrections are controlled.	Payroll Administrator	Payroll Officer	F3.13	Should
US-3.20	As Finance, I want run totals and the Bank + Cash = Total check so that the run reconciles.	Finance / Accounts	Finance Reviewer	F3.14	Must

8. Module 4: Bonus Management

8.1 Purpose

Calculate bonus on gross salary at any time of the year. Bonus is set up here as a **Bonus Type: a name with a percentage**, and it is processed in its own bonus run. The bank bonus is 80% of the bonus and the rest is paid as cash bonus. Bonus percentages belong to this module and not to Payroll Management.

Dependencies: Module 3 for the gross (Total) Salary effective on the bonus date, and the ERP for employee status. The module does not need a payroll run, so bonus can run in any month.

8.2 Features

ID	Feature	Functionality	Primary actor
F4.1	Bonus Type Setup	Create bonus types as a name with a percentage of gross salary (for example Festival Bonus, 60%) and a bank portion % (default 80%). Effective-dated; Management approves; inactive types cannot be used.	HR Officer, Management Approver
F4.2	Bonus run creation	Create a bonus run at any time of the year: choose Bonus Type, occasion name and bonus date; include all eligible employees or select some. Any number of runs per year.	Payroll Officer
F4.3	Gross salary fetch	Read each employee's gross (Total) Salary from the payroll profile effective on the bonus date and store it in the run. Only active ERP employees with a Complete profile are included.	Payroll System
F4.4	Bonus calculation	Total Bonus = Gross × Bonus %; Bank Bonus = Total Bonus × 80%; Cash Bonus = the rest. Rounding as per parameters; Bank + Cash always equals Total.	Payroll System
F4.5	Exclusions and validation	Exclude an employee with a reason; warn on a duplicate for the same Bonus Type and occasion; validate missing gross salary and the Bank + Cash check.	Payroll Officer
F4.6	Review, approval and finalization	Draft, Reviewed (Finance), Approved (Management), Finalized (locked). Preparer cannot approve.	Finance Reviewer, Management Approver, Payroll Officer
F4.7	Bonus register and export	Register of bonus runs with totals split into bank and cash for payment; Excel and PDF export (also in Module 8).	Payroll Officer, Finance Reviewer

8.3 Business Logic

> **Bonus calculation**
> **Total Bonus** = Gross (Total) Salary × Bonus %
> **Bank Bonus** = Total Bonus × Bank portion % (80%)
> **Cash Bonus** = Total Bonus − Bank Bonus (the rest, in hand cash)

- Bonus setup (name with percentage and bank portion) is owned by this module and stored with each run (BL-20).
- A bonus run can be created at any time and does not depend on a payroll run; the gross salary is taken from the profile version effective on the bonus date (BL-21).

- Bank Bonus is rounded and Cash Bonus takes the remainder, so the two always add up to Total Bonus (BL-22).
- The run follows Draft, Reviewed, Approved, Finalized; the preparer cannot approve; finalization locks it (BL-23).
- Salary components, Bank Salary and Cash Salary of the payroll run are not used.

8.4 Process Flow

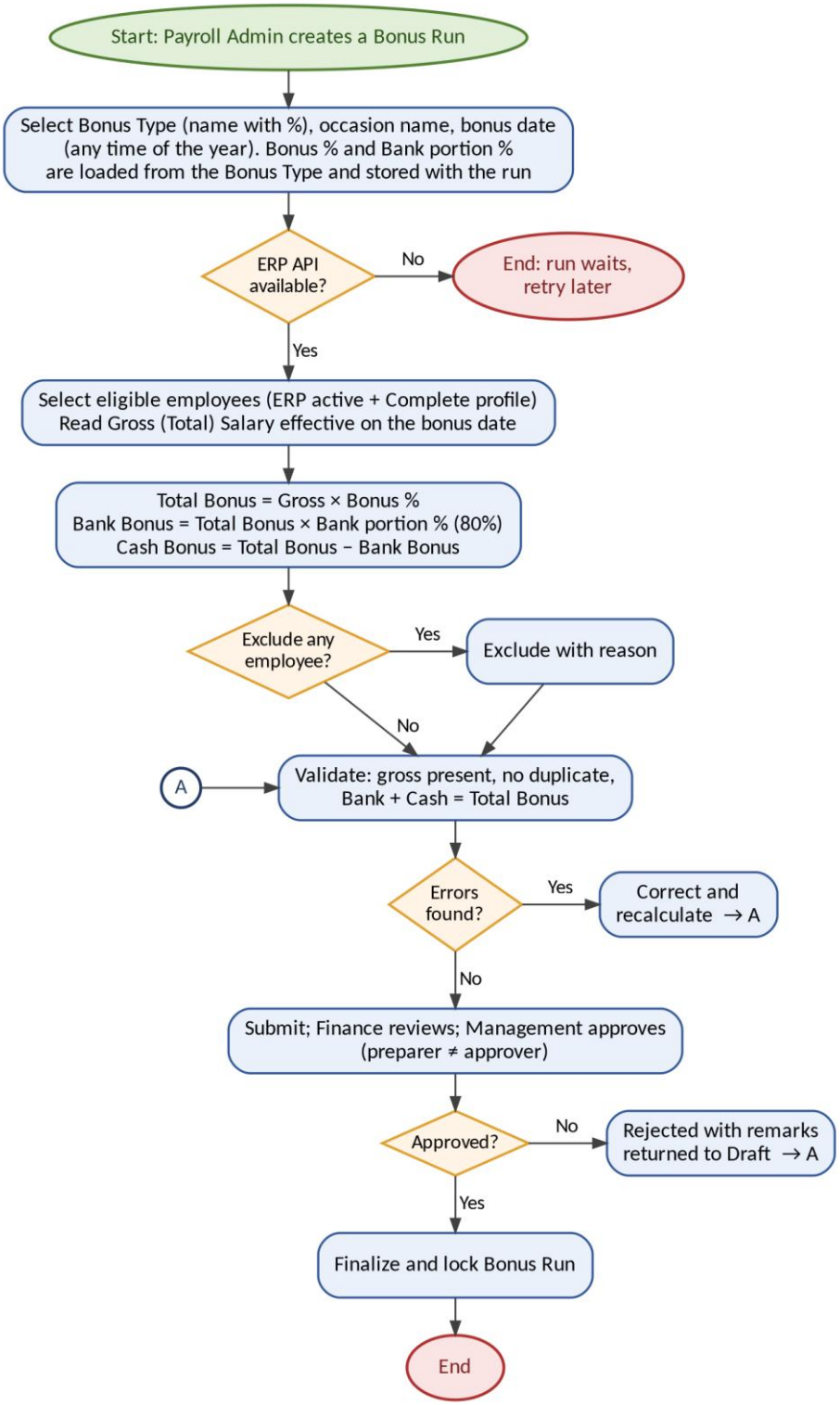


Figure 6: Bonus processing flow (any time of the year)

8.5 UI Screen Specification

Screen 4.1: Bonus Type List and Form

Role: HR Administrator (Management approves) | Menu: Bonus > Bonus Types

Field / Control	Type	Options / Format / Behaviour	Mandatory
Bonus Type Name	Text (60)	Unique, for example Festival Bonus.	Yes
Bonus Percentage	Decimal	Percent of gross salary, 0 to 100.	Yes
Bank Portion %	Decimal	Default 80; the rest is cash bonus.	Yes
Effective From / To	Date	No overlap for the same name.	Yes / No
Approval Status	Badge	Draft / Approved; only Approved types can be used.	N/A
Status	Toggle	Active / Inactive.	Yes

Screen 4.2: Bonus Run List and Create

Role: Payroll Administrator | Menu: Bonus > Bonus Run

Field / Control	Type	Options / Format / Behaviour	Mandatory
Bonus Run No	Auto text	BN-YYYY-NNN.	N/A
Bonus Type	Dropdown	Active, approved Bonus Types; shows the percentage.	Yes
Occasion Name	Text (60)	For example Eid 2026.	Yes
Bonus Date	Date	Any date; gross salary is taken as at this date.	Yes
Employees	Radio + multi-select	All eligible / selected employees.	Yes
Status / Version	Badge / Read-only	Draft, Reviewed, Approved, Finalized, Rejected.	N/A
Create Run	Button	Builds lines; blocked with a message if the ERP is unavailable.	N/A

Screen 4.3: Bonus Run Detail

Role: Payroll Administrator, Finance, Management (by status) | Opens from Screen 4.2

Field / Control	Type	Options / Format / Behaviour	Mandatory
Employee ID, Name	Read-only	From the ERP, stored with the line.	N/A
Gross (Total) Salary	Read-only decimal	Effective on the bonus date.	N/A
Bonus %, Bank Portion %	Read-only decimal	From the Bonus Type, stored with the run.	N/A
Total Bonus, Bank Bonus, Cash Bonus	Read-only decimal	BL-22.	N/A
Exclude Employee	Checkbox + reason	Optional; reason required when used.	Reason if used
Flags	Icon list	Missing gross salary, duplicate, check difference.	N/A
Totals Row	Read-only	Total, bank and cash bonus; headcount.	N/A
Validate / Submit / Review / Approve / Reject / Finalize	Buttons	Same roles as the payroll run; Reject needs remarks.	N/A

Field / Control	Type	Options / Format / Behaviour	Mandatory
Export	Button	Excel or PDF bonus register.	N/A

8.6 User Stories

ID	User story	Role	Actor	Feature	Priority
US-4.1	As an HR Administrator, I want to set up a bonus as a name with a percentage so that each bonus type is configured, not hard-coded.	HR Administrator	HR Officer	F4.1	Must
US-4.2	As Management, I want to approve bonus types before use so that bonus policy is controlled.	Management	Management Approver	F4.1	Must
US-4.3	As a Payroll Administrator, I want to create a bonus run at any time of the year so that bonus is not tied to the salary run.	Payroll Administrator	Payroll Officer	F4.2	Must
US-4.4	As the Payroll System, I need to take each employee's gross salary as at the bonus date so that the bonus uses the right basis.	System (automated)	Payroll System	F4.3	Must
US-4.5	As Finance, I want the bonus split into 80% bank bonus and the rest cash bonus so that payment can be prepared for bank and cash.	Finance / Accounts	Finance Reviewer	F4.4	Must
US-4.6	As a Payroll Administrator, I want to exclude an employee with a reason and see duplicate warnings so that the run is accurate.	Payroll Administrator	Payroll Officer	F4.5	Should
US-4.7	As Finance, I want to review and reject a bonus run with remarks so that errors are caught.	Finance / Accounts	Finance Reviewer	F4.6	Must
US-4.8	As Management, I want to approve a bonus run that I did not prepare so that duties are separated.	Management	Management Approver	F4.6	Must
US-4.9	As a Payroll Administrator, I want to finalize a bonus run and export the bank and cash lists so that the bonus can be paid.	Payroll Administrator	Payroll Officer	F4.6, F4.7	Must

9. Module 5: Payslip Management

9.1 Purpose

Generate, print and keep **Bank Payslips and Cash Payslips** for every finalized salary run. Payslips are produced together with the salary generation and are maintained here. The module reads only the finalized run snapshot from Module 3 and owns its own templates.

9.2 Features

ID	Feature	Functionality	Primary actor
F5.1	Payslip template setup	Maintain a Bank Payslip template and a Cash Payslip template: organization header, footer note, which heads to show, whether Total Salary shows on the Bank Payslip (default off), amount in words, acknowledgement line on the Cash Payslip, numbering format.	HR Officer
F5.2	Payslip generation	Generate payslips automatically when a run is finalized (event Run Finalized). A Bank Payslip for each employee with Bank Salary above 0 and a Cash Payslip for each employee with Cash Salary above 0. A missed run can be generated manually.	Payroll System, Payroll Officer
F5.3	Print and download	Print or download a payslip for one employee, selected employees or the whole run as one combined PDF; choose Bank, Cash or both. Every reprint is logged.	Payroll Officer, HR Officer
F5.4	Payslip register	List payslips by run, employee, type and status; open and download.	Payroll Officer, HR Officer, Finance Reviewer
F5.5	Supersede on re-open	When a run is re-opened and finalized again, earlier payslips become Superseded and are kept; new payslips are generated.	Payroll System
F5.6	Privacy control	Each payslip holds only its own employee's data and only one salary type; access follows roles.	Payroll System

9.3 Business Logic

- Payslips come only from a finalized payroll run and its stored snapshot, never from current setups (BL-24).
- Generation is triggered by the event Run Finalized. If this module is unavailable, the run is still finalized and payslips are generated when it returns (BL-24).
- Bank Payslip shows bank components only; Cash Payslip shows Cash Salary only (BL-25).
- Numbers are unique (BPS/YYYY/NNNNN and CPS/YYYY/NNNNN); payslips are immutable; superseded ones stay on file (BL-24, BL-26).

Bank Payslip content

Section	Content
Header	Organization name and logo; payslip number; payroll period; run number; issue date.
Employee	Employee ID, name, designation (from the ERP, as stored with the run), tier.
Earnings	Basic and every head flagged "show on payslip" (for example H/R, Conveyance, M/A), each with its amount.
Total	**Bank Salary** in figures and in words. Total Salary shows only if the template allows it.
Footer	Note text from the template; "System generated".

Cash Payslip content

Section	Content
Header	Organization name and logo; payslip number; payroll period; run number; issue date.
Employee	Employee ID, name, designation, tier.
Amount	**Cash Salary** in figures and in words. No bank components are shown.
Acknowledgement	Received by (signature) and date.
Footer	Note text from the template.

9.4 Process Flow

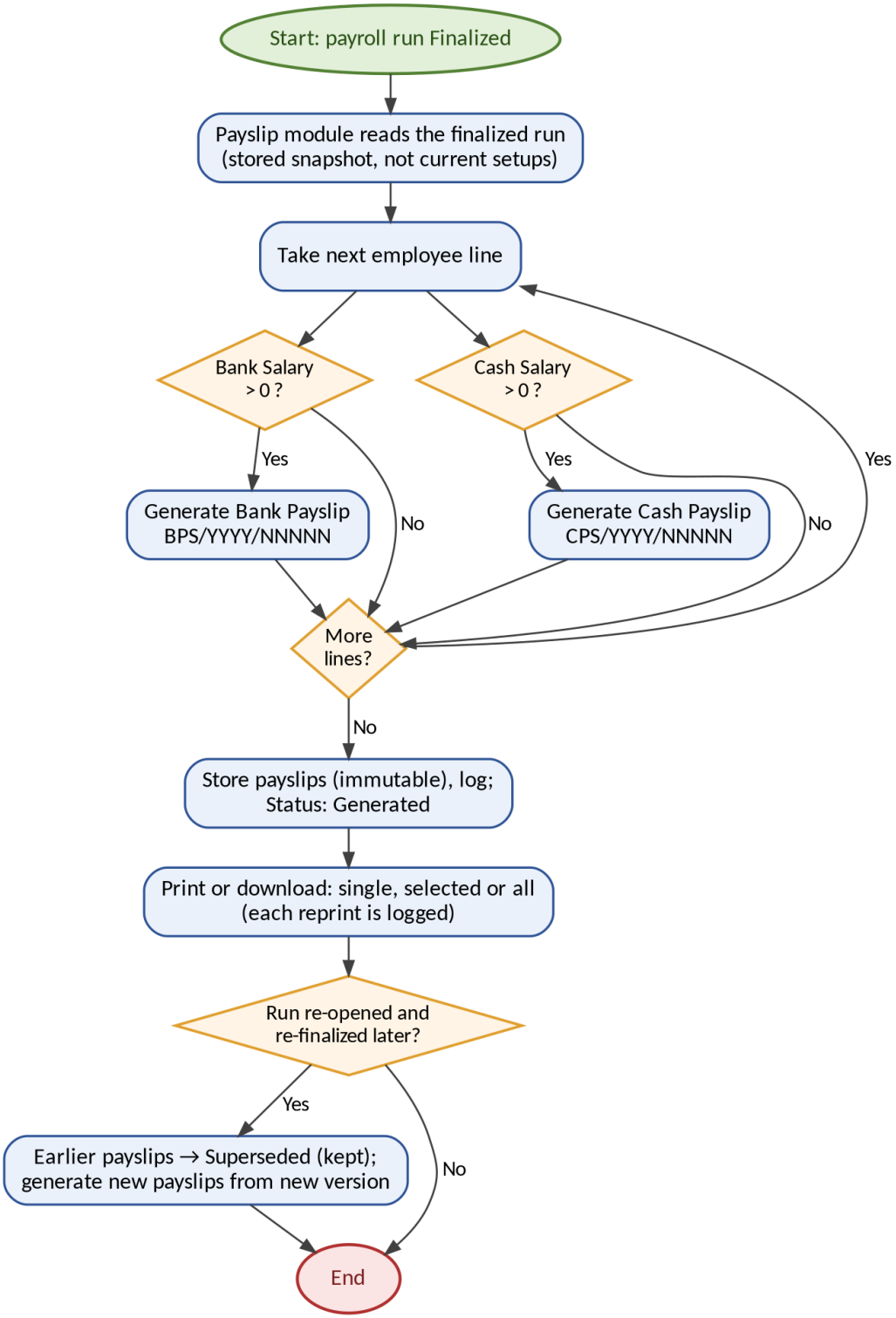


Figure 7: Payslip generation and printing flow

9.5 UI Screen Specification

Screen 5.1: Payslip Template Setup

Role: HR Administrator | Menu: Payslip > Payslip Templates

Field / Control	Type	Options / Format / Behaviour	Mandatory
Template Type	Dropdown	Bank Payslip / Cash Payslip.	Yes
Header Text and Logo	Text / Image	Organization header.	Yes
Heads to Show	Multi-select	Active heads flagged "show on payslip"; Bank Payslip only.	Conditional
Show Total Salary	Checkbox	Bank Payslip only; default off.	No
Show Amount in Words	Checkbox	Default on.	No
Acknowledgement Line	Checkbox	Cash Payslip only; default on.	No
Footer Note	Text (250)	Free text.	No
Numbering Format	Text	BPS/{YYYY}/{NNNNN} or CPS/{YYYY}/{NNNNN}; sequence resets yearly.	Yes
Status	Toggle	Active / Inactive.	Yes

Screen 5.2: Generate and Print Payslips

Role: Payroll Administrator, HR Administrator | Menu: Payslip > Generate and Print Payslips

Field / Control	Type	Options / Format / Behaviour	Mandatory
Finalized Payroll Run	Dropdown	Finalized runs only.	Yes
Payslip Type	Radio	Bank / Cash / Both.	Yes
Employees	Radio + multi-select	All / selected.	Yes
Generation Status	Read-only	Generated automatically, or Not generated with a Generate button.	N/A
Preview	Button	Shows the payslip layout.	N/A
Print / Download PDF	Buttons	Single, selected or combined PDF; each action is logged.	N/A

Screen 5.3: Payslip Register

Role: Payroll Administrator, HR Administrator, Finance (view) | Menu: Payslip > Payslip Register

Field / Control	Type	Options / Format / Behaviour	Mandatory
Payslip No	Read-only	Unique number.	N/A
Employee ID / Name	Read-only		N/A
Type, Period, Run No	Read-only	Bank or Cash.	N/A
Amount	Read-only	Bank Salary or Cash Salary.	N/A
Status	Badge	Generated / Superseded.	N/A
Download / Reprint	Buttons	Every reprint is logged.	N/A
Search and filter	Text + dropdown	By number, employee, run, type, status, period.	No

9.6 User Stories

ID	User story	Role	Actor	Feature	Priority
US-5.1	As an HR Administrator, I want to design the Bank and Cash payslip templates so that payslips carry the approved layout.	HR Administrator	HR Officer	F5.1	Should
US-5.2	As the Payroll System, I need to generate Bank and Cash payslips when a run is finalized so that payslips are produced with the salary.	System (automated)	Payroll System	F5.2	Must
US-5.3	As a Payroll Administrator, I want to generate payslips manually for a finalized run that was missed so that no employee is left without a payslip.	Payroll Administrator	Payroll Officer	F5.2	Should
US-5.4	As a Payroll Administrator, I want to print or download payslips for one, some or all employees so that I can hand them out.	Payroll Administrator	Payroll Officer	F5.3	Must
US-5.5	As an HR Administrator, I want to find and reprint an employee's payslip from the register so that I can answer employee requests.	HR Administrator	HR Officer	F5.3, F5.4	Must
US-5.6	As Finance, I want to view the payslip register so that I can check issued payslips against the run.	Finance / Accounts	Finance Reviewer	F5.4	Could
US-5.7	As the Payroll System, I need to mark earlier payslips Superseded when a re-opened run is finalized again so that only current payslips are in use.	System (automated)	Payroll System	F5.5	Should
US-5.8	As an HR Administrator, I want the Bank Payslip to hide cash details and the Cash Payslip to hide bank details so that each slip shows only what it should.	HR Administrator	HR Officer	F5.6	Must

10. Module 6: Salary Certificate Request Approval

10.1 Purpose

The **HR Approval Desk**. An employee requests a salary certificate in the ERP; the request reaches Payroll through the API; HR verifies it and approves or rejects it; the status and the approved certificate return to the ERP, where the employee downloads it. HR can also download the certificate from Payroll while verifying, before approving.

This module owns only the request and the HR decision. It does not build certificates: it asks Module 7 to issue or preview one. Module 7 knows nothing about requests, so the two modules stay loosely coupled.

10.2 Features

ID	Feature	Functionality	Primary actor
F6.1	ERP request intake	Receive the employee's request from the ERP through the API: ERP request ID, employee ID, purpose, addressee, request date. A request ID already received is rejected as a duplicate. Status = Requested.	ERP System, Payroll System
F6.2	Approval queue	List of requests with status, filters (status, date, employee, purpose) and ageing of pending requests.	HR Officer
F6.3	Request verification	Open a request and see the employee (from the ERP), purpose, addressee and payroll availability. The system checks that the employee is active and has finalized payroll.	HR Officer
F6.4	Approve or reject	Approve the request, or reject it with a mandatory reason. The approver cannot be the requester. On approval the certificate is issued by Module 7.	HR Officer
F6.5	HR working copy download	HR can download the certificate PDF at any time during verification, without approving. The copy is marked "Not approved" and has no number.	HR Officer
F6.6	Status and certificate to the ERP	Return the status (Requested, Approved, Rejected with reason) and, after approval, the certificate PDF to the ERP through the API so that the employee can download it. Keep the request history.	Payroll System, Employee

10.3 Business Logic

- Requests originate only in the ERP, on request of the employee (BL-27).
- Every request is verified by HR; approval requires an active employee with finalized payroll; rejection requires a reason; the approver cannot be the requester (BL-28).
- Before approval HR may download a working copy only; the employee can download only after approval (BL-29).
- On approval, Module 7 issues the certificate and its status and PDF become available to the ERP (BL-30).
- If the ERP is unavailable, the decision is saved and the status is sent when the ERP is reachable again; no decision is lost (BL-01).

> **Request status**
> **Requested**: received from the ERP, waiting for HR.
> **Approved**: HR approved; certificate issued and shown with its number.
> **Rejected**: HR rejected with a reason; the reason is returned to the ERP.

10.4 Process Flow

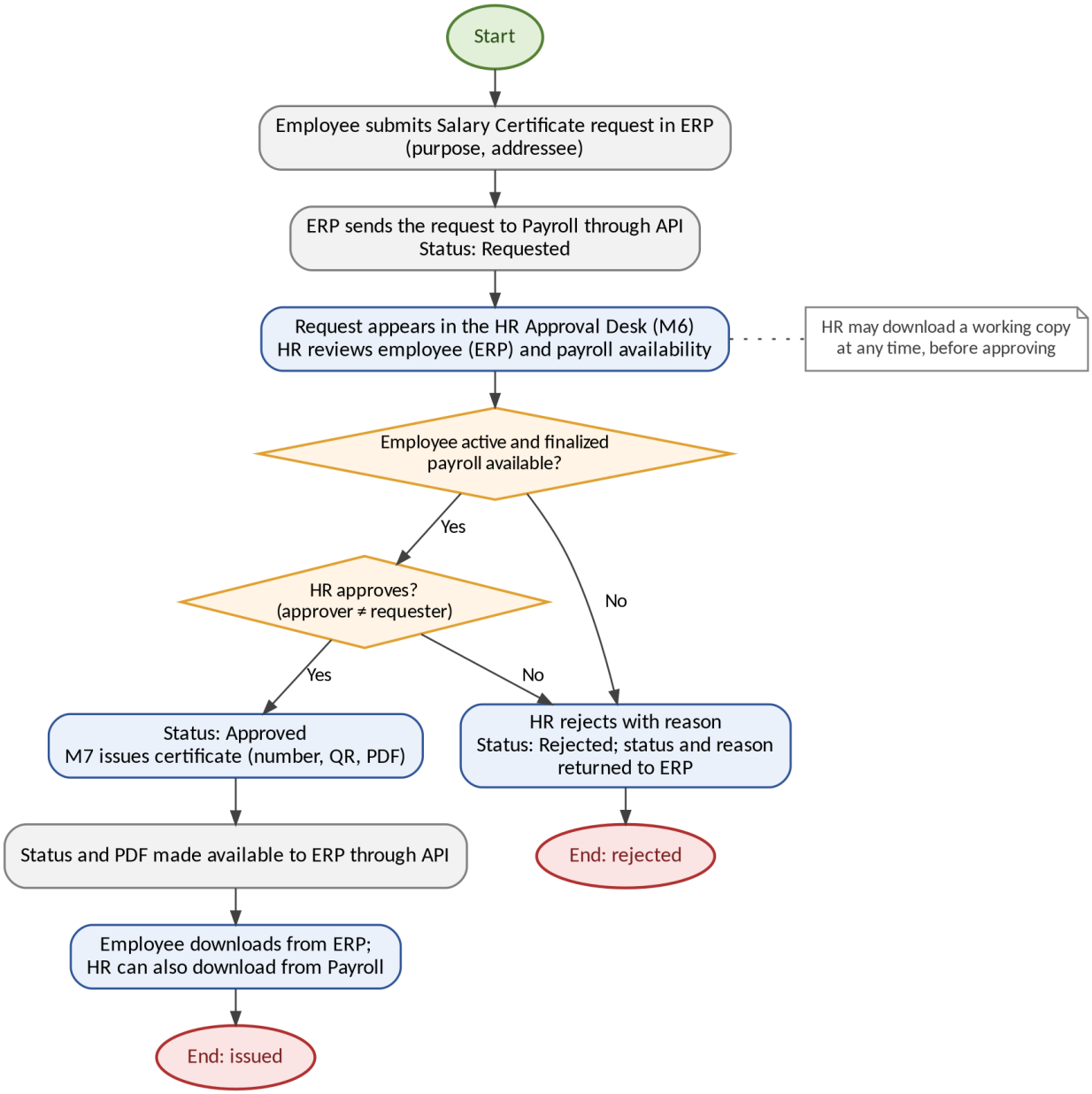


Figure 8: Salary certificate request and HR approval flow

10.5 UI Screen Specification

Screen 6.1: Approval Queue

Role: HR Administrator | Menu: Salary Certificate > Approval Desk

Field / Control	Type	Options / Format / Behaviour	Mandatory
Request ID (ERP)	Read-only	ERP request number.	N/A
Employee ID / Name	Read-only	From the ERP.	N/A
Purpose / Addressee	Read-only	As entered by the employee.	N/A
Requested On / Age	Read-only	Days waiting.	N/A
Status	Badge	Requested / Approved / Rejected.	N/A

Field / Control	Type	Options / Format / Behaviour	Mandatory
Filter	Dropdown + date	Status, date range, employee, purpose.	No
Open	Button	Opens Screen 6.2.	N/A

Screen 6.2: Request Verification and Approval

Role: HR Administrator | Opens from Screen 6.1

Field / Control	Type	Options / Format / Behaviour	Mandatory
Employee ID, Name, Designation, Joining Date	Read-only	Fetched from the ERP.	N/A
Request details	Read-only	Request ID, purpose, addressee, request date.	N/A
Eligibility check	Read-only	Active in the ERP: yes / no. Finalized payroll available: yes / no (with period).	N/A
Template	Dropdown	Active certificate templates (Module 7); default by purpose.	Yes
Preview / Download Working Copy	Buttons	Available at any time before approval; copy marked "Not approved", no number.	N/A
Approve	Button	Enabled only when eligibility passes and the user is not the requester. Issues the certificate.	N/A
Reject	Button	Opens a reason box.	N/A
Rejection Reason	Text (250)	Returned to the ERP.	Reason: Yes
Certificate No / Download	Read-only / Button	After approval: number and the issued PDF.	N/A

Screen 6.3: Request History

Role: HR Administrator | Opens from Screen 6.2

Field / Control	Type	Options / Format / Behaviour	Mandatory
Event	Read-only	Received, Opened, Working copy downloaded, Approved, Rejected, Status sent to ERP.	N/A
User, Date and Time, Remarks	Read-only	From the audit log.	N/A

Screen 6.4: Employee Request (ERP side, reference only)

Role: Employee in the ERP | Built and maintained in the ERP; listed so that the interface is clear

Field / Control	Type	Options / Format / Behaviour	Mandatory
Employee	Read-only	The logged-in employee.	N/A
Purpose	Dropdown + text	Bank, Visa, Embassy, General, Other (text).	Yes
Addressee	Text (150)	Defaults to "To Whom It May Concern".	No
Submit	Button	Sends the request to Payroll through the API.	N/A
My Requests	Grid	Request, date, status (Requested / Approved / Rejected), rejection reason.	N/A
Download Certificate	Button	Enabled only when Approved; fetches the PDF from Payroll through the API.	N/A

10.6 User Stories

ID	User story	Role	Actor	Feature	Priority
US-6.1	As an Employee, I want to request a salary certificate in the ERP so that I do not have to ask HR in person.	ERP Employee	Employee (in ERP)	F6.1	Must
US-6.2	As the Payroll System, I need to receive the ERP request and reject a duplicate request ID so that each request appears once.	System (automated)	Payroll System	F6.1	Must
US-6.3	As an HR Administrator, I want a queue of pending requests with ageing so that nothing waits unnoticed.	HR Administrator	HR Officer	F6.2	Must
US-6.4	As an HR Administrator, I want to see the employee, purpose and payroll availability so that I can verify the request.	HR Administrator	HR Officer	F6.3	Must
US-6.5	As an HR Administrator, I want to approve a request so that the certificate is issued and the employee can download it.	HR Administrator	HR Officer	F6.4	Must
US-6.6	As an HR Administrator, I want to reject a request with a reason so that the employee knows why.	HR Administrator	HR Officer	F6.4	Must
US-6.7	As an HR Administrator, I want to download a working copy before approving so that I can check the certificate without waiting.	HR Administrator	HR Officer	F6.5	Must
US-6.8	As an Employee, I want to see the status in the ERP and download the approved certificate there so that I get it without visiting HR.	ERP Employee	Employee (in ERP)	F6.6	Must
US-6.9	As Management, I want HR to be unable to approve a request raised by the same person so that approvals are independent.	Management	Management Approver	F6.4	Should

11. Module 7: Salary Certificate Issuance

11.1 Purpose

Produce the official salary certificate: template, numbering, QR verification code, PDF, register, download, cancel and re-issue. Module 7 is a document service. It issues a certificate when Module 6 reports an approved request, or when HR issues one directly. It reads only the finalized payroll snapshot from Module 3 and the employee data from the ERP.

11.2 Features

ID	Feature	Functionality	Primary actor
F7.1	Certificate template setup	Maintain templates: body text with placeholders, figures to print (Total Salary only, breakdown with Total Salary, or Bank Salary only), amount in words, signatory, signature image, numbering format.	HR Officer, Management Approver
F7.2	Issue and preview service	Called by Module 6: preview (working copy, no number) or issue (number, QR, PDF) for an eligible employee. Figures come from the latest finalized payroll.	Payroll System
F7.3	HR direct issue	HR can issue a certificate directly, with no ERP request and no approval, for an eligible employee. Source is recorded as "HR direct".	HR Officer
F7.4	Register, download and reprint	Register of certificates with number, employee, template, purpose, source, dates and status; download and reprint (each reprint logged).	HR Officer
F7.5	Cancel and re-issue	Cancel an issued certificate with a mandatory reason; the number stays as Cancelled; issue a new one.	HR Officer
F7.6	QR verification page	A public page opened by the QR code shows the certificate number, employee name, issue date and Valid or Cancelled.	Third party

11.3 Business Logic

- A certificate can be issued only for an active employee with finalized payroll (BL-30).
- Figures come only from the latest finalized payroll; the template decides what prints; Cash Salary and the 80% amount do not print by default (BL-30).
- Number format SC/YYYY/NNNN, reset yearly; each certificate has a QR code; an issued certificate is immutable (BL-30, BL-32).
- HR direct issue needs no request and no approval and follows the same rules (BL-31).
- A change needs Cancel and Re-issue with a reason; cancelled numbers stay on file; reprints are logged (BL-32).
- Name, designation and joining date are stored with the certificate (BL-03).

Certificate content (template placeholders)

Section	Content
Header	Organization name and logo, certificate number, issue date.
Addressee	{Addressee} or To Whom It May Concern.
Body	This is to certify that {Employee Name} (Employee ID {Employee ID}) is employed with {Organization} as {Designation} since {Joining Date}. The present monthly salary structure of the employee is stated below.

Section	Content
Salary table	Per template: Basic and heads flagged "show on certificate", Total Salary; amount in figures and in words ({Currency}).
Purpose line	This certificate is issued on request for {Purpose} and carries no liability for the organization.
Signature	{Signatory Name}, {Signatory Designation}, signature image.
Footer	QR verification code and verification text.

Wording is a starting point and must be approved by HR and Management. Pronouns are avoided because gender is not an ERP field in scope.

11.4 Process Flow

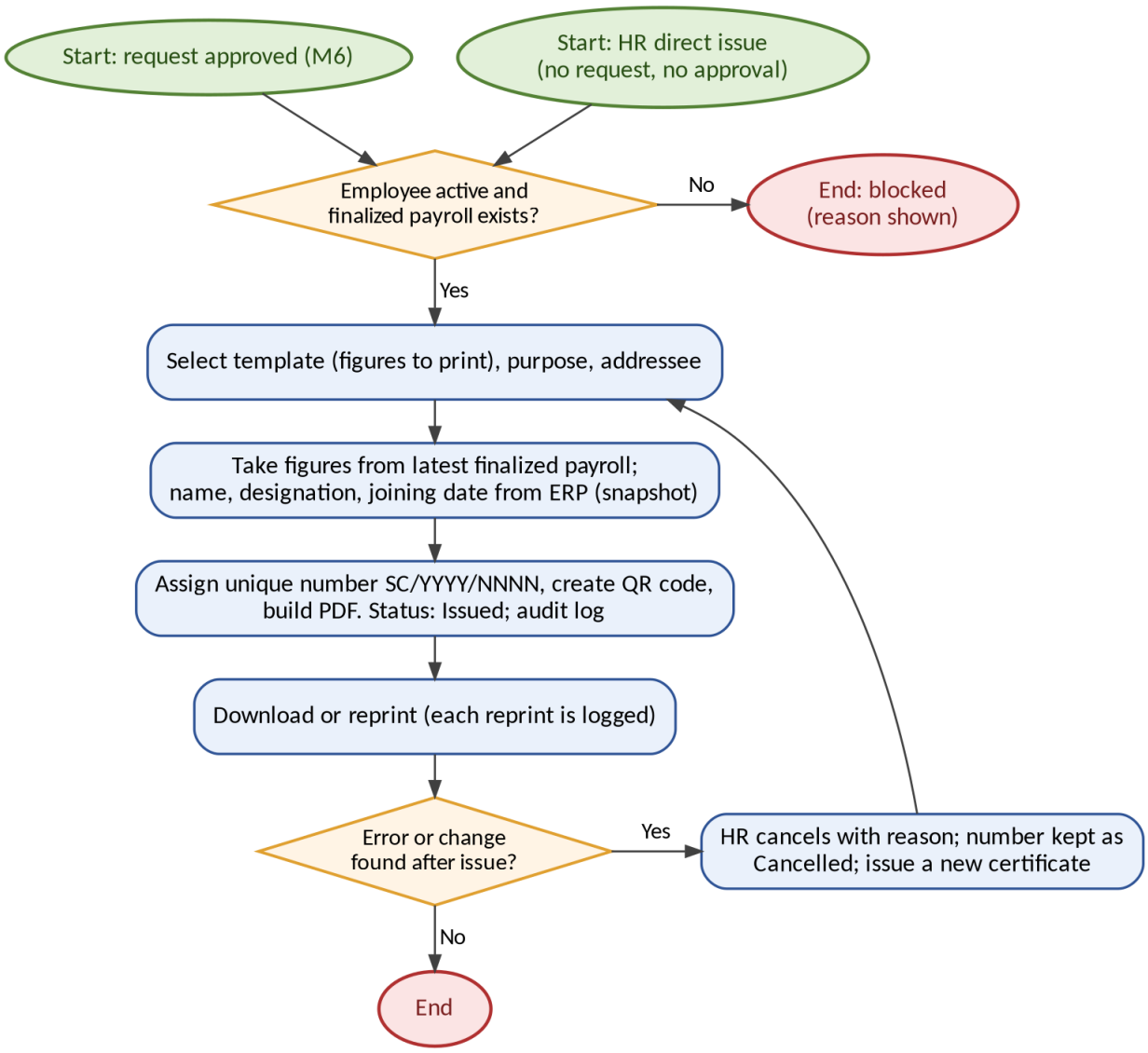


Figure 9: Salary certificate issuance, download and cancel / re-issue flow

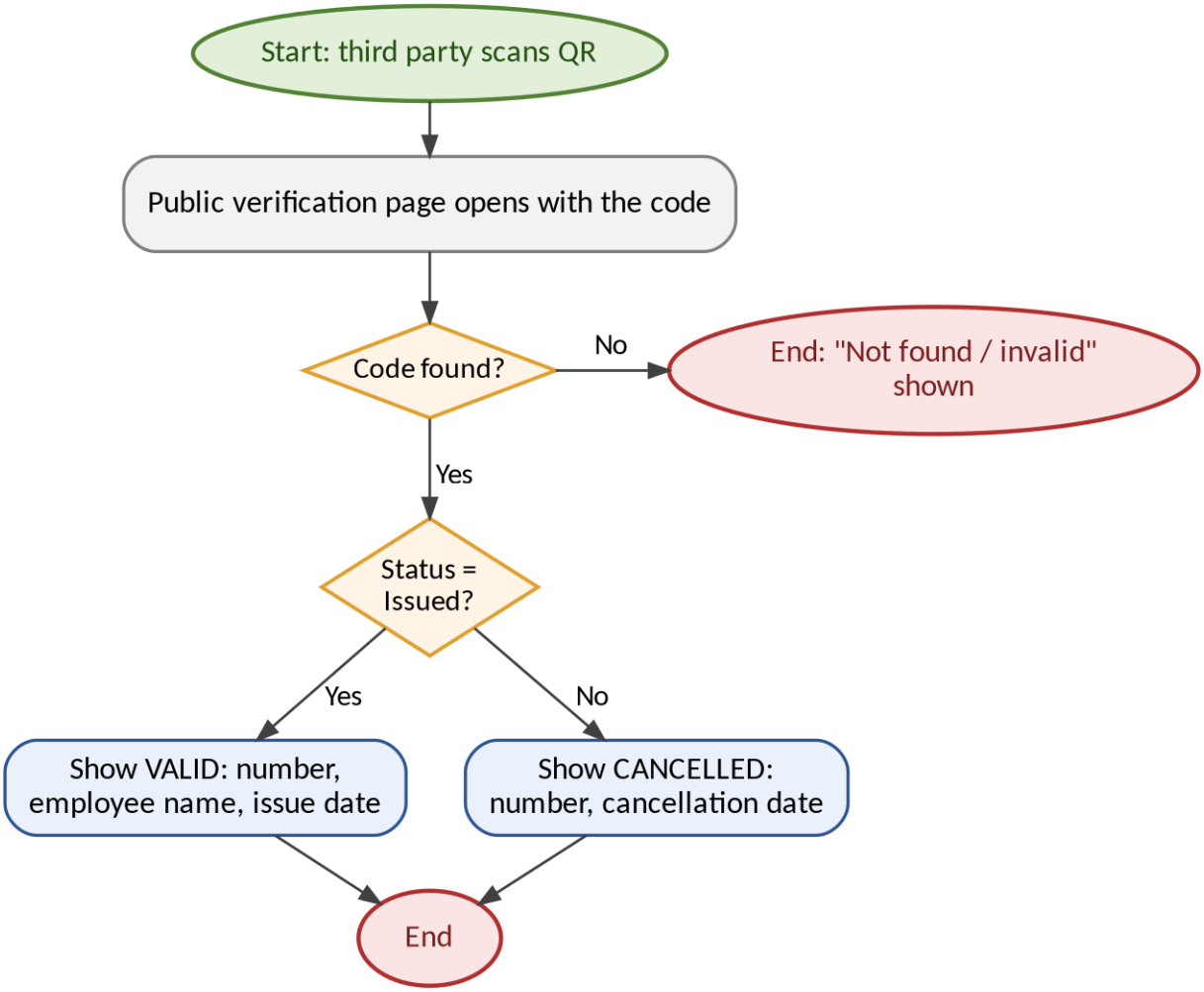


Figure 10: QR verification flow

11.5 UI Screen Specification

Screen 7.1: Certificate Template Setup

Role: HR Administrator (Management approves wording) | Menu: Salary Certificate > Certificate Templates

Field / Control	Type	Options / Format / Behaviour	Mandatory
Template Name	Text (60)	Unique, for example General, Bank, Visa.	Yes
Body Text	Rich text	With the placeholders listed above.	Yes
Salary Figures to Show	Multi-select	Total Salary, Basic, heads flagged "show on certificate", Bank Salary.	Yes
Show Amount in Words	Checkbox	Default on.	No
Signatory Name and Designation	Text	Authorized signatory.	Yes
Signature Image	Image upload	PNG or JPG, size limit applies.	No
Numbering Format	Text	SC/{YYYY}/{NNNN}; sequence resets yearly.	Yes
Approval Status / Status	Badge / Toggle	Draft / Approved; Active / Inactive.	Yes

Screen 7.2: Issue Certificate (HR direct)

Role: HR Administrator | Menu: Salary Certificate > Issue Certificate

Field / Control	Type	Options / Format / Behaviour	Mandatory
Employee	Searchable dropdown	Active ERP employees with finalized payroll; shows ID and name.	Yes
Template	Dropdown	Active, approved templates.	Yes
Purpose	Dropdown + text	Bank, Visa, Embassy, General, Other (text).	Yes
Addressee	Text (150)	Defaults to To Whom It May Concern.	No
Salary As Of	Read-only	Period of the latest finalized run used.	N/A
Preview	Button	Shows the filled certificate; unsaved.	N/A
Issue and Download	Button	No approval needed; assigns number and QR.	N/A

Screen 7.3: Certificate Register

Role: HR Administrator, Management (view) | Menu: Salary Certificate > Certificate Register

Field / Control	Type	Options / Format / Behaviour	Mandatory
Certificate No	Read-only	Unique number.	N/A
Employee ID / Name	Read-only		N/A
Template, Purpose, Addressee	Read-only		N/A
Source	Read-only	ERP request (with request ID) / HR direct.	N/A
Issued By / Date	Read-only	From the audit log.	N/A
Status	Badge	Issued / Cancelled.	N/A
Download PDF / Reprint	Buttons	Issued only; every reprint is logged.	N/A
Cancel and Re-issue	Button	Reason mandatory.	Reason: Yes
Search and filter	Text + dropdown	By number, employee, status, source, date range.	No

Screen 7.4: Public Verification Page

Role: any person with the QR code (no login) | Opened by scanning the QR

Field / Control	Type	Options / Format / Behaviour	Mandatory
Certificate No	Read-only	From the QR code.	N/A
Employee Name	Read-only	As stored with the certificate.	N/A
Issue Date	Read-only		N/A
Result	Badge	VALID / CANCELLED / Not found. No salary figures are shown.	N/A

11.6 User Stories

ID	User story	Role	Actor	Feature	Priority
US-7.1	As an HR Administrator, I want to design certificate templates so that each purpose has approved wording and figures.	HR Administrator	HR Officer	F7.1	Must
US-7.2	As Management, I want to approve certificate wording and the signatory so that certificates are authorized.	Management	Management Approver	F7.1	Should
US-7.3	As the Payroll System, I need to issue a numbered certificate with QR and PDF when HR approves a request so that the employee can download it.	System (automated)	Payroll System	F7.2	Must
US-7.4	As an HR Administrator, I want to issue a certificate directly without an ERP request or approval so that urgent cases are handled quickly.	HR Administrator	HR Officer	F7.3	Must
US-7.5	As an HR Administrator, I want to search, download and reprint certificates from a register so that I can answer repeat requests.	HR Administrator	HR Officer	F7.4	Must
US-7.6	As an HR Administrator, I want to cancel and re-issue a certificate with a reason so that mistakes are corrected without losing the old number.	HR Administrator	HR Officer	F7.5	Should
US-7.7	As a bank or embassy officer, I want to scan the QR code and see whether the certificate is valid so that I can trust it.	System (automated)	Third party (no login)	F7.6	Should

12. Module 8: Reports

12.1 Purpose

Reports read from the other modules through read-only interfaces and own no business data. Reports on finalized runs, bonus runs, payslips and certificates read the stored values and snapshots, not current setups.

12.2 Features

ID	Feature	Functionality	Primary actor
F8.1	Report catalogue	The reports listed in Section 12.3, each with filters and a defined layout.	All report users
F8.2	Filters and export	Filter by run, period, employee, tier, status or date; export to Excel and PDF; every export is logged.	All report users
F8.3	Report access control	Each report is limited to the roles allowed to see its data (BL-34).	System Admin

12.3 Report Catalogue

Report	Content	Source	Filters	Export
Employee Payroll Detail	Employee data, tier, scale, N, Basic, heads, Bank, Cash, Total, 80% amount, variance, formula breakdown.	M3	Run, employee, tier	Excel, PDF
Payroll Summary	Headcount; totals of scales, increments, Basic, heads, Bank, Cash, Total, 80% amount.	M3	Run; group by tier	Excel, PDF
Payroll Register	Employee-level values of a finalized run in the approved layout.	M3	Run	Excel, PDF
Bank Salary Sheet / Cash Salary Sheet	Employees with Bank Salary / Cash Salary and totals, for payment preparation.	M3	Run	Excel, PDF
Tuning Variance Report	Bank vs 80% amount per employee, suggested N vs chosen N, override reasons.	M3	Run, variance range	Excel
Exception and Validation Report	Missing fields, negative cash, reconciliation errors, overrides.	M3	Run, severity	Excel, PDF
Bonus Register	Gross, total bonus, bank bonus and cash bonus per employee and totals.	M4	Bonus run, type	Excel, PDF
Payslip Register	Payslips issued: number, employee, type, run, status, reprints.	M5	Run, type, status	Excel, PDF
Certificate Request Status	Requests with status, ageing, approver, rejection reason.	M6	Status, date	Excel, PDF
Salary Certificate Register	Numbers issued, status, purpose, source, issuer.	M7	Date, employee, status	Excel, PDF
Setup Change History	Tier, scale, head, parameter and bonus type changes with approver and dates.	M1 to M4	Setup, date	Excel
Setup Usage Report	Tiers, scales and heads in use by employees.	M1 to M3	Setup	Excel
Audit Report	Who changed what and when.	Common	Date, user, entity	Excel

12.4 Report Rules

- Reports on finalized records read the stored values and snapshots (BL-07).
- Salary data is shown only to roles with permission; every export is logged (BL-34).
- Bank account details and other sensitive fields are not included until the business confirms the need (Chapter 19).
- Export layouts for Excel and PDF are approved by Finance.

12.5 User Stories

ID	User story	Role	Actor	Feature	Priority
US-8.1	As Finance, I want payroll summary, register and bank / cash sheets so that I can verify and prepare payments.	Finance / Accounts	Finance Reviewer	F8.1	Must
US-8.2	As Management, I want bonus, variance and exception reports so that I can control cost and exceptions.	Management	Management Approver	F8.1	Should
US-8.3	As an HR Administrator, I want certificate and request reports so that I can follow pending and issued certificates.	HR Administrator	HR Officer	F8.1	Should
US-8.4	As a System Administrator, I want an audit report so that I can trace changes.	System Administrator	System Admin	F8.1	Must
US-8.5	As a Payroll Administrator, I want to export reports to Excel and PDF so that I can share them.	Payroll Administrator	Payroll Officer	F8.2	Must

13. User Story Map: Role and Actor

Every user story is mapped to the role that holds the permission and the actor that performs it. The tables below are built from the story lists in the module chapters.

13.1 Stories by Role

Role	Actors	Stories	Count
HR Administrator	HR Officer	US-1.1, US-1.2, US-1.4, US-1.6, US-2.1, US-2.2, US-2.3, US-2.5, US-3.24, US-4.1, US-5.1, US-5.5, US-5.8, US-6.3, US-6.4, US-6.5, US-6.6, US-6.7, US-7.1, US-7.4, US-7.5, US-7.6, US-8.3	23
Payroll Administrator	Payroll Officer	US-3.1, US-3.2, US-3.4, US-3.5, US-3.7, US-3.8, US-3.9, US-3.21, US-3.10, US-3.11, US-3.13, US-3.14, US-3.15, US-3.18, US-3.19, US-4.3, US-4.6, US-4.9, US-5.3, US-5.4, US-8.5	21
Finance / Accounts	Finance Reviewer	US-3.12, US-3.16, US-3.20, US-4.5, US-4.7, US-5.6, US-8.1	7
Management	Management Approver	US-1.3, US-2.4, US-3.3, US-3.22, US-3.17, US-4.2, US-4.8, US-6.9, US-7.2, US-8.2	10
System Administrator	System Admin	US-8.4	1
ERP Employee	Employee (in ERP)	US-6.1, US-6.8	2
System (automated)	Payroll System, Third party (no login)	US-1.5, US-3.6, US-3.23, US-4.4, US-5.2, US-5.7, US-6.2, US-7.3, US-7.7	9

13.2 Stories by Actor

Actor	Role	Stories	Count
HR Officer	HR Administrator	US-1.1, US-1.2, US-1.4, US-1.6, US-2.1, US-2.2, US-2.3, US-2.5, US-3.24, US-4.1, US-5.1, US-5.5, US-5.8, US-6.3, US-6.4, US-6.5, US-6.6, US-6.7, US-7.1, US-7.4, US-7.5, US-7.6, US-8.3	23
Management Approver	Management	US-1.3, US-2.4, US-3.3, US-3.22, US-3.17, US-4.2, US-4.8, US-6.9, US-7.2, US-8.2	10
Payroll System	System (automated)	US-1.5, US-3.6, US-3.23, US-4.4, US-5.2, US-5.7, US-6.2, US-7.3	8
Payroll Officer	Payroll Administrator	US-3.1, US-3.2, US-3.4, US-3.5, US-3.7, US-3.8, US-3.9, US-3.21, US-3.10, US-3.11, US-3.13, US-3.14, US-3.15, US-3.18, US-3.19, US-4.3, US-4.6, US-4.9, US-5.3, US-5.4, US-8.5	21
Finance Reviewer	Finance / Accounts	US-3.12, US-3.16, US-3.20, US-4.5, US-4.7, US-5.6, US-8.1	7
Employee (in ERP)	ERP Employee	US-6.1, US-6.8	2
Third party (no login)	System (automated)	US-7.7	1
System Admin	System Administrator	US-8.4	1

13.3 Stories by Module

Module	Stories	Must	Total
Module 1 Tier of Employee Setup	US-1.1 to US-1.6	4	6
Module 2 Scale Setup	US-2.1 to US-2.5	3	5
Module 3 Payroll Management	US-3.1 to US-3.20	19	24
Module 4 Bonus Management	US-4.1 to US-4.9	8	9

Module	Stories	Must	Total
Module 5 Payslip Management	US-5.1 to US-5.8	4	8
Module 6 Certificate Request Approval	US-6.1 to US-6.9	8	9
Module 7 Certificate Issuance	US-7.1 to US-7.7	4	7
Module 8 Reports	US-8.1 to US-8.5	3	5

14. System-Wide Controls (Common Services)

Common Services are shared by all modules. They are services, not business modules, and they hold no business rules of their own.

14.1 Common Services

Service	Function	Used by
Audit trail	Records who did what and when, with old and new values.	All modules
Access control	Role-based permission on every screen, report and export.	All modules
Number sequences	Run numbers, bonus run numbers, payslip numbers, certificate numbers, with yearly reset where defined.	M3, M4, M5, M7
PDF and QR generation	PDF with Unicode fonts, combined PDF, QR code, number-to-words.	M5, M7
Event notification	Run Finalized and Request Approved events delivered to the interested module.	M3, M5, M6, M7

14.2 Audit Trail

- Log every create, edit, status change, approval, finalization, re-open, override, payslip print and certificate action with user, date and time, old value and new value (BL-33).
- Audit records cannot be edited or deleted by any role; the System Administrator can view and export them.
- Retention period to be defined by the business (Chapter 19).

14.3 Security and Privacy

- Role-based access on every screen, report and export (BL-34); salary fields hidden from roles without permission.
- Login by company credentials or ERP single sign-on, per IT decision; session timeout; encrypted transport.
- The ERP and Payroll APIs use authenticated, encrypted calls.
- Personal and salary data is protected from unauthorized access and is not shown in logs or error messages.
- The public verification page shows no salary figures.

14.4 Data Integrity

- Money values are stored as fixed-point decimals, never floating point.
- Employee ID uniqueness, valid dates and non-negative whole-number increment counts are enforced.
- Calculated fields are read-only; editable fields are clearly marked (BL-15).
- Finalized runs, bonus runs, payslips and issued certificates are immutable.

14.5 Non-Functional Requirements

Area	Requirement
Performance	Run calculation, bulk payslip generation and reports complete within an agreed time for the expected headcount (53 employees in the ERP sample); target to be confirmed.
Modularity	A module can be deployed, changed or switched off without code change in another module; cross-module access is read-only and one-directional.
Availability and backup	Backup frequency, recovery objectives and retention defined by IT.

Area	Requirement
Usability	Formulas and component breakdowns visible; clear separation of input and calculated fields; Phase 2 menu items visibly disabled.
Deployment	Web, desktop or other model to be confirmed by IT.
Compatibility	Modern browsers; PDF and Excel outputs open in standard tools.
Maintainability	Heads, rates, percentages, templates and number formats configurable without code change (BL-08, BL-35).

15. Dependencies

15.1 ERP Integration (API): Payroll calls the ERP

- ERP API: get an employee by ID, search employees, fetch several employees at once
- ERP employee fields: Employee ID, name, phone number, joining date, designation, department, employment status
- ERP API: list of designations
- ERP API authentication method (key, token or SSO)

15.2 Payroll Interface: the ERP calls Payroll

- Payroll API: submit a salary certificate request (ERP request ID, employee ID, purpose, addressee, request date)
- Payroll API: get the status of a request (Requested, Approved, Rejected with reason)
- Payroll API: download the approved certificate PDF
- ERP screens for the employee request and download (Screen 6.4)

15.3 Other Dependencies

- Network connection between the payroll system and the ERP
- Clean ERP master data: unique IDs, valid joining dates, phone numbers in one format, clearly defined designations
- Login method for payroll users (own login or SSO)
- Relational database with decimal support
- Hosting and deployment environment
- PDF generation with Unicode fonts, including combined PDF for bulk payslips
- Excel import and export library
- Number-to-words and date formatting (currency aware)
- QR code generation and a public verification page
- Backup and restore facility
- Approved business rules and sign-off
- Email or SMS gateway (Phase 2)

16. Data Migration and Baseline Reconciliation

16.1 Migration Steps

1. Check the ERP API data for all employees; correct duplicate IDs, phone numbers, dates and designations with the ERP team.
2. Create scales from the worksheet: 9000-800, 10000-900, 12000-1000, 15000-1100.
3. Create the tiers (for example SM, Senior), link them to scales and map ERP designations to tiers.
4. Load the salary heads and rates: H/R 50%, M/A 10%, Conveyance 1,000; and the payroll parameters (Bank target 80%, rounding, currency, period type).
5. Create Bonus Types (name with percentage) once the business confirms them.
6. Map each worksheet row to an ERP employee ID (names repeat in the worksheet, so mapping is by confirmation, not by name).
7. Import Total Salary and N from the worksheet; the system recalculates every component.
8. Compare the system totals with the baseline below; investigate any difference before go-live.

16.2 Baseline Totals (18 Employees)

Component	Baseline total
Starting salary scales	181,000
Total increment amount	159,400
Basic salary	340,400
House rent allowance	170,200
Conveyance allowance	18,000
Medical allowance	34,040
Bank salary	562,640
Cash salary	130,860
Total salary	693,500
80% of total salary	554,800

Data issues to resolve before migration

(1) The worksheet has 19 rows, one more than the 18-employee baseline.

(2) The 19th row (Himal) has a blank reconciliation cell and a check value of 10,000.

(3) Several rows exceed the 80% amount by a large margin (rows 12, 14 and 16 are 2,920, 5,800 and 1,960 above); migrate them as confirmed overrides with a reason.

17. Acceptance Criteria

ID	Scenario	Expected result	Rule
AT-01	Scale 9,000, increment 800, N = 2	Total increment 1,600; Basic 10,600.	BL-09
AT-02	Basic 10,600; H/R 50%, Conveyance 1,000, M/A 10%	H/R 5,300; M/A 1,060; Bank 17,960.	BL-08, 11
AT-03	Total 24,000; Bank 17,960	Cash 6,040; Bank + Cash = 24,000.	BL-11
AT-04	Total 24,000	80% amount 19,200; Bank and Cash unchanged by it.	BL-12
AT-05	Bank greater than Total	Negative cash flagged; finalization blocked unless override with reason.	BL-11
AT-06	Baseline of 18 sample employees	Totals equal Chapter 16 baseline (Total 693,500; Bank 562,640; Cash 130,860).	BL-11
AT-07	Scale 9000-800, Total 24,000	Suggested N = 3; Bank 19,240; variance +40; N = 2 accepted as override with reason.	BL-13
AT-08	Scale 9000-800, Total 20,000	N = 0; Bank 15,400; Cash 4,600; flag Basic at scale minimum.	BL-13
AT-09	Scale 10000-900, Total 70,000	Suggested N = 27; Bank 55,880; Cash 14,120.	BL-13
AT-10	N entered as 2.5 or −1; N above maximum	Rejected; whole number 0 or more, not above maximum.	BL-09
AT-11	ERP returns an invalid or missing joining date, or a duplicate ID	Error shown; employee blocked until the ERP data is corrected.	BL-01
AT-12	ERP returns phone 1717212912	Shown as 01717212912.	BL-01
AT-13	ERP returns the employee as inactive	Excluded from new runs, bonus runs and certificates; earlier records kept.	BL-02
AT-14	ERP API unavailable	Message shown; retry possible; no payroll data changed.	BL-01
AT-15	Add a new head (for example Special Allowance, 5% of Basic, part of Bank Salary)	Included in the calculation and on the Bank Payslip with no code change.	BL-08
AT-16	Head rate changed from a future date	Finalized run values and rule snapshot unchanged; new rate used from its date after approval.	BL-07
AT-17	Tier created and linked to a scale; employee assigned the tier	Employee inherits the scale; designation shown from the ERP.	BL-04, 06
AT-18	ERP designation mapped to a tier	Tier suggested in the profile; HR can change it.	BL-04
AT-19	Scale override without a reason	Rejected; with a reason it is saved and audited.	BL-06
AT-20	Assign an inactive tier	Not allowed.	BL-04
AT-21	Preparer tries to approve own run or bonus run	Action unavailable.	BL-18, 23
AT-22	Finalized run, then scale or percentage changes	Finalized run values and rule snapshot intact.	BL-07, 19
AT-23	Re-open request on a finalized run	Needs approval; new version in Draft; earlier version kept.	BL-19
AT-24	Second run for the same period	Rejected while another run is active.	BL-16
AT-25	Bonus: Gross 24,000, Bonus 60%, bank portion 80%	Total bonus 14,400; bank bonus 11,520; cash bonus 2,880.	BL-22
AT-26	Bonus: Gross 10,001, 60%, 80%	Total 6,001; bank 4,801; cash 1,200; bank + cash = total.	BL-22, 14
AT-27	Create a bonus run in a month with no payroll run	Allowed; gross salary taken as at the bonus date.	BL-21
AT-28	Bonus Type percentage changed after a bonus run is finalized	Finalized bonus run unchanged.	BL-20, 07

ID	Scenario	Expected result	Rule
AT-29	Exclude an employee from a bonus run	Reason required; totals recalculated; duplicate warning for same type and occasion.	BL-23
AT-30	Run finalized	Bank Payslip (Basic, heads, Bank 17,960) and Cash Payslip (Cash 6,040) generated automatically with unique numbers.	BL-24
AT-31	Employee with Cash Salary 0	No Cash Payslip generated.	BL-24
AT-32	Bank Payslip opened	No cash amount shown; Cash Payslip shows no bank components.	BL-25
AT-33	Run re-opened and finalized again	Earlier payslips Superseded and kept; new payslips generated.	BL-26
AT-34	Payslip module unavailable at finalization	Run is finalized; payslips generated when the module returns.	BL-24
AT-35	Employee submits a request in the ERP	Request appears in the HR Approval Desk as Requested; a repeated request ID is rejected.	BL-27
AT-36	HR approves a request	Certificate issued with number, QR and PDF; status and PDF available in the ERP; employee downloads it.	BL-28, 30
AT-37	HR rejects a request	Reason mandatory; status Rejected and reason shown in the ERP.	BL-28
AT-38	HR downloads before approving	Working copy marked "Not approved", no number; employee cannot download yet.	BL-29
AT-39	HR user approves a request raised by the same person	Not allowed.	BL-28
AT-40	HR issues a certificate directly	Numbered certificate issued with no approval; source "HR direct".	BL-31
AT-41	Certificate for employee without finalized payroll	Blocked with reason.	BL-30
AT-42	Cancel and re-issue	Old number stays Cancelled; QR shows Cancelled; new number issued.	BL-32
AT-43	Scan QR of an issued certificate	Page shows number, name, issue date and VALID.	BL-32
AT-44	User without permission opens a salary report	Access denied; attempt logged.	BL-34
AT-45	Setup edited or override recorded	Audit entry with user, time, old and new value.	BL-33
AT-46	Total 24,000; increment 2,000 entered as a separate input	Incremented Total Salary 26,000; initial 24,000 kept in history; suggested N = 4, Bank 20,520, Cash 5,480.	BL-10, 13
AT-47	Increment saved with an effective date after a finalized run	Finalized run unchanged; new total used from the next period; bonus run dated after the effective date uses 26,000.	BL-10, 21
AT-48	Increment amount 0 or negative; increment not yet approved	Rejected; an unapproved increment does not change Total Salary.	BL-10

18. Phase 2: Future Scope

Area	Description
Deductions and net pay	Income tax and statutory deductions, loans, advances; net-pay calculation.
Attendance, leave, overtime	Integration with the ERP or attendance devices to drive payable days and overtime. Menu placeholders are visible in Phase 1 (Section 3.4).
Bonus enhancements	Performance-rating-based bonus rules, arrears, pro-rata for new joiners, multi-level bonus approval.
Payslip delivery and self-service	Payslip email or SMS delivery and viewing in the ERP employee portal; bonus payslips if required.
Bank and cash disbursement	Bank advice and bulk transfer file; cash disbursement sheet.
Accounting integration	Payroll journal posting to the ERP general ledger.
Increment and promotion workflow	Effective-dated salary revision history with approval.
Benefits	Provident fund, gratuity and other benefits.
Approvals	Multi-level, role-based approval workflows.
Analytics	Dashboards: cost by department, trend, variance.
ERP write-back and multi-company	Return payroll status to the ERP; support several companies.
Certificate enhancements	Digital signature, public verification portal, email notification of status.
Mobile and notifications	Mobile access, email and SMS alerts.

19. Open Items and Decisions Required

Each item carries the default used in this BRD. A default is not an agreed business rule until the business confirms it.

No	Question	Default in this BRD	Owner	Affects
1	May Bank Salary exceed the 80% amount, or is "closest" the only rule? The worksheet shows both.	Closest wins; user may override with reason.	Management, Finance	BL-13
2	What is the purpose of the 80% figure and is it a fixed policy?	Configurable Bank target %, 80%.	Management	BL-12
3	Is Conveyance 1,000 for everyone, now and later? Which other heads will be added?	Fixed head in setup, editable with approval.	HR	BL-08
4	Rounding method and unit.	Half-up, whole unit.	Finance	BL-14
5	Negative Cash Salary: warn or block?	Block, with authorized override.	Finance	BL-11
6	Which tiers exist (for example SM, Senior)? Is the scale decided by tier with employee-level override? Is there a maximum increment count?	By tier, override allowed; maximum optional.	HR	BL-04, 06
7	The review suggested mapping the ERP designation to the scale. Map designation to tier and tier to scale (this BRD), or directly designation to scale? Are the ERP designations clearly defined?	Designation to tier to scale.	HR, ERP team	BL-04
8	Which figures print on the salary certificate, and who signs?	Breakdown with Total Salary; signatory set in template.	HR, Management	BL-30
9	Bonus Types: which names and percentages? Is 60% of gross still the rate?	Configurable per Bonus Type; 60% illustrative.	Management	BL-20
10	Should the bonus bank portion (80%) be set per Bonus Type or follow the payroll Bank target %?	Per Bonus Type, default 80%.	Management, Finance	BL-20, 22
11	Bonus eligibility: minimum service, pro-rata for new joiners?	All eligible employees; exclusion with reason; no pro-rata.	Management, HR	BL-21, 23
12	Currency, payroll frequency and period naming.	Configurable; not assumed.	Finance	BL-35
13	ERP API details: endpoints, authentication, response fields (including designation, department, status), response time.	To be provided by the ERP team.	ERP team	Ch 15
14	How does the certificate request reach Payroll: the ERP sends it (default) or Payroll fetches pending requests?	ERP calls the Payroll API.	ERP team	BL-27
15	Who is the HR approver, and may several HR users approve? Can an HR user approve his or her own request?	HR Administrator role; own request not allowed.	HR, Management	BL-28
16	HR download before approval: working copy marked "Not approved" (default), or a normal copy?	Working copy, no number.	HR, Management	BL-29
17	Issue the certificate automatically on approval (default) or as a separate step?	Automatic on approval.	HR	BL-30
18	Employee download in the ERP: PDF served by Payroll on demand (default), or stored in the ERP?	Served by Payroll through the API.	ERP team	BL-30
19	Who may cancel a certificate, and is the employee notified in the ERP?	HR with reason; no notification in Phase 1.	HR	BL-32

No	Question	Default in this BRD	Owner	Affects
20	Payslip content: which heads print, Total Salary on the Bank Payslip, acknowledgement line on the Cash Payslip?	Heads flagged; Total off; acknowledgement on.	HR, Finance	BL-25
21	Are payslips needed for bonus (bank bonus and cash bonus) in Phase 1?	No; salary payslips only.	Management, HR	BL-24
22	Payslip timing: automatic on finalization (default) or manual trigger?	Automatic; manual regenerate available.	Payroll Admin	BL-24
23	The 19th worksheet row (Himal) and the 18-row baseline.	Treated as data issue until confirmed.	Payroll Admin	Ch 16
24	Audit and payroll history retention period.	To be defined.	Management, IT	Ch 14
25	Variance tolerance for warnings.	5% of the 80% amount.	Finance	BL-17
26	Is separation of preparer and approver acceptable for a small team?	Required.	Management	BL-18, 23
27	Should reports include bank account details?	No.	Finance	M8
28	Does a salary increment need Management approval, and is a mid-period effective date paid pro-rata?	Management approves; applies from the next full period, no pro-rata.	Management, Finance	BL-10

20. Conclusion and Sign-off

This BRD restructures the Payroll Management System into eight independent modules that follow the review comments: Tier of Employee setup with the designation taken from the ERP, salary head and bonus percentage setups kept inside their own modules, Bank and Cash payslips and bonus in Phase 1, and a salary certificate flow that starts in the ERP, is approved by HR and is downloadable from both systems. The calculation rules come from the worksheet; the items marked Proposed and the questions in Chapter 19 need business decisions before design and build.

Sign-off confirms that the documented requirements are suitable for software design. Unresolved items stay open and must not be treated as agreed business rules.

Approval Item	Name / Role	Decision / Notes	Date
Business rules approved by			
HR requirements reviewed by			
Finance / Accounts requirements reviewed by			
ERP integration reviewed by			
Software scope approved by			
Version / change reference			